Crash Course in Mathematics

for Cognitive Science

Lecturer Notes

Probability & Statistics · Linear Algebra

Five 3-hour sessions preparing students for cognitive modeling, computational neuroscience, machine learning, and data analysis.

Master of Cognitive Science — École Normale Supérieure, Paris

How to use these notes

These notes are written for the *lecturer*. They contain the full narrative of each session: the intuition to convey, the derivations worth showing on the board, worked examples with complete solutions, and explicit links to cognitive science. Everything is designed for students who may have *no* mathematical background, so the guiding principle throughout is **intuition first, formalism second**.

The colour code. Boxes are used consistently so you can navigate a session at a glance:

Definition	the precise statement, after the intuition.
Key idea	the one thing students must remember.
Intuition	the mental picture, in plain words.
Worked example	to solve on the board, with full solution.
Cognitive-science link	where the concept pays off.
Common pitfall	the mistake students reliably make.
Recap	the closing summary of the session.

The five sessions.

1. Probability I — the language of uncertainty (no background assumed).
2. Probability & Statistics II — from data to inference (no background).
3. Linear Algebra I — vectors, matrices, and geometry (no background).
4. Linear Algebra II — eigenvectors, PCA, and dynamics (no background).
5. Advanced Probability & Statistics — the modeller's toolkit (recap).

Timing markers (*e.g.* "$\approx$*25 min*") suggest a pace that fits each session into three hours including a short break and the in-class exercises, which are provided in a separate booklet.

Contents

Chapter 1

Probability I — The Language of Uncertainty

Where this session is going

By the end of today students should be able to: describe an uncertain situation with a *sample space* and *events*; compute probabilities using the basic rules; update a belief with *conditional probability* and *Bayes' theorem*; and recognise the handful of *distributions* (Bernoulli, Binomial, Poisson, Gaussian) that describe most phenomena they will meet in modeling and neuroscience. The single most important idea of the day is **Bayes' theorem**: how evidence changes belief.

1.1 Why probability? (≈*15 min*)

Almost everything a cognitive scientist studies is *uncertain*. A subject's response on the next trial, the number of spikes a neuron will fire, whether a classifier will label an image correctly — none of these are fixed in advance. Probability is simply the mathematics of reasoning carefully about such uncertainty. It gives us a language to say *how likely* something is, and rules to combine those likelihoods without contradicting ourselves.

Intuition

Think of a probability as a number between 0 and 1 that measures our *degree of certainty*. Zero means "impossible", one means "certain", and 0.5 means "as likely as not". Everything in this session is built from this one scale and a few common-sense rules for combining numbers on it.

Why it matters for cognitive science

A dominant view in modern cognitive science is that the brain itself is, in part, a probabilistic inference machine: perception is treated as inferring the most probable cause of ambiguous sensory data, and learning as updating beliefs from experience. The tools in this session are literally the vocabulary of that theory — *Bayesian models of cognition*, *ideal-observer analysis*, and the probabilistic models underlying machine learning all start here.

1.2 Sample spaces and events (≈*25 min*)

Before we can talk about how likely something is, we must be precise about *what can happen.*

Definition

The **sample space** Ω is the set of all possible outcomes of an experiment. An **event** A is a subset of Ω: a collection of outcomes we care about.

For a single die roll, $\Omega = \{1, 2, 3, 4, 5, 6\}$ and "the result is even" is the event $A = \{2, 4, 6\}$. For a subject's response in a yes/no detection task, $\Omega = \{\text{yes}, \text{no}\}$. For the number of spikes in one second, $\Omega = \{0, 1, 2, 3, \dots\}$.

Intuition

Events are just *questions with a yes/no answer* about the outcome. "Is the die even?" carves Ω into the outcomes where the answer is yes and those where it is no. All the set operations below are just ways of building new yes/no questions from old ones.

Because events are sets, we combine them with set operations, which correspond to everyday logical words:

- $A \cup B$ (**union**, "A *or* B"): outcomes in A, in B, or in both.
- $A \cap B$ (**intersection**, "A *and* B"): outcomes in both.
- A^c (**complement**, "*not* A"): outcomes not in A.
- A and B are **disjoint** (mutually exclusive) if $A \cap B = \varnothing$: they cannot both happen.

1.2.1 Counting outcomes

When every outcome is equally likely, probability reduces to *counting*: $\mathbb{P}(A) = \dfrac{\#\text{outcomes in } A}{\#\text{outcomes in } \Omega}$. So we need to count efficiently.

Key idea

Two rules cover almost all counting we need.

Permutations (order matters): the number of ordered arrangements of k items chosen from n is $\dfrac{n!}{(n-k)!}$.

Combinations (order does not matter): the number of unordered subsets of size k from n is $\dbinom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$.

Counting committees

A lab has 8 members. How many ways can we choose a 3-person reading group?

Solution. Order does not matter (a group is a set), so we use a combination: $\binom{8}{3} = \dfrac{8!}{3!\,5!} = \dfrac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56$.

Lecturer note: Emphasise why *we divide by $3!$: the $8 \cdot 7 \cdot 6 = 336$ ordered choices each count the same group $3! = 6$ times.*

Three counting facts worth memorising

1. **Multiplication principle:** if a first choice has a options and a second (independent) choice has b options, together they have $a \cdot b$.
2. $\binom{n}{k} = \binom{n}{n-k}$ (choosing which k to *include* is the same as choosing which $n-k$ to *leave out*).
3. $\binom{n}{0} = \binom{n}{n} = 1$ and $\binom{n}{1} = n$.

Order matters: a podium

Eight sprinters race; how many different gold–silver–bronze podiums are possible?

Solution. Order matters (gold $\neq$ silver), so this is a permutation: $\dfrac{8!}{(8-3)!} = 8 \cdot 7 \cdot 6 = 336$. Compare with the 56 unordered reading groups above: the podium counts each trio in all $3! = 6$ orders.

Counting — do together

1. A questionnaire has 5 yes/no items. How many distinct response patterns are possible?
   Answer. $2^5 = 32$ (multiplication principle).
2. From a group of 10 participants, how many ways to form a 4-person focus group?
   Answer. $\binom{10}{4} = 210$.
3. In how many orders can 6 pictures be shown to a subject?
   Answer. $6! = 720$.

1.3 The rules of probability (≈*20 min*)

We now attach numbers to events. A valid assignment of probabilities must obey three simple axioms (Kolmogorov's axioms), which just formalise common sense.

Definition

A **probability** is a function $\mathbb{P}$ assigning to each event a number, such that

1. $\mathbb{P}(A) \geq 0$ for every event A;
2. $\mathbb{P}(\Omega) = 1$ (something must happen);
3. if A and B are disjoint, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.

From these, everything else follows. The three you will use constantly:

Key idea

Complement rule: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
Addition rule: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
Range: $0 \leq \mathbb{P}(A) \leq 1$ for every event.

The addition rule's subtraction is the whole story: if we add $\mathbb{P}(A)$ and $\mathbb{P}(B)$ we count the overlap $A \cap B$ twice, so we remove it once.

Or, and, not

In a study, 60% of participants are bilingual (B) and 30% are musicians (M); 20% are both. Pick a participant at random.

Solution. $\mathbb{P}(\text{bilingual or musician}) = \mathbb{P}(B \cup M) = 0.60 + 0.30 - 0.20 = 0.70$.
$\mathbb{P}(\text{neither}) = 1 - 0.70 = 0.30$ by the complement rule.
$\mathbb{P}(\text{bilingual but not musician}) = \mathbb{P}(B) - \mathbb{P}(B \cap M) = 0.60 - 0.20 = 0.40$.

Common pitfall

The addition rule only reduces to $\mathbb{P}(A)+\mathbb{P}(B)$ *when the events are disjoint.* Students routinely forget the $-\mathbb{P}(A \cap B)$ term and get probabilities greater than 1. A probability above 1 is always a signal that something was double-counted.

Rules that follow from the three axioms

For any events A, B:

- **Impossible event:** $\mathbb{P}(\varnothing) = 0$.
- **Monotonicity:** if $A \subseteq B$ then $\mathbb{P}(A) \le \mathbb{P}(B)$ (a smaller event cannot be more likely).
- **Complement:** $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
- **Inclusion–exclusion (three events):**

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C).$$

Intuition

The “−overlaps+triple” pattern of inclusion–exclusion is just careful bookkeeping: adding the three singles double-counts every pairwise overlap, so we subtract them; but the triple overlap, added three times and subtracted three times, has vanished, so we add it back once. For everyday problems the *complement rule* is the workhorse — “at least one” is almost always easiest as $1 - \mathbb{P}(\text{none})$.

“At least one” via the complement

A shape appears on each of 3 independent trials with probability 0.2 of being a target. What is the probability of *at least one* target across the 3 trials?

Solution. Going through “at least one” directly is messy; use the complement. The probability of *no* target on a single trial is 0.8, so across three independent trials $\mathbb{P}(\text{no target}) = 0.8^3 = 0.512$. Hence $\mathbb{P}(\text{at least one}) = 1 - 0.512 = 0.488$.

Lecturer note: This “$1 - (\text{none})$” trick recurs all term; drill it now.

Probability rules — do together

In a sample, $\mathbb{P}(\text{coffee}) = 0.5$, $\mathbb{P}(\text{tea}) = 0.3$, $\mathbb{P}(\text{both}) = 0.1$.

1. $\mathbb{P}(\text{coffee or tea})$?

   Answer. $0.5 + 0.3 - 0.1 = 0.7$.
2. $\mathbb{P}(\text{neither})$?

   Answer. $1 - 0.7 = 0.3$.
3. $\mathbb{P}(\text{exactly one of the two})$?

   Answer. $0.7 - 0.1 = 0.6$ (union minus the intersection).

1.4 Conditional probability and independence (≈*30 min*)

This is the conceptual heart of the session: *how does knowing one thing change the probability of another?*

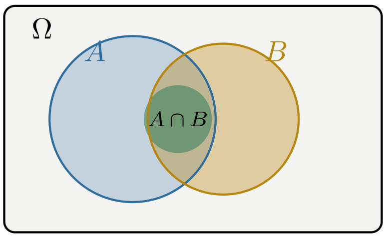
Definition

The **conditional probability** of A given B (assuming $\mathbb{P}(B) > 0$) is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Intuition

Conditioning on B means *zooming in*: we throw away every outcome where B did not happen, and ask what fraction of the remaining world also has A. The denominator $\mathbb{P}(B)$ renormalises so the shrunken world again has total probability 1.



Conditioning on B:
the world becomes the B circle,
and $\mathbb{P}(A \mid B)$ is the
fraction of it covered by A.

Rearranging the definition gives the **multiplication rule**, which lets us build the probability of a conjunction step by step:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\,\mathbb{P}(B) = \mathbb{P}(B \mid A)\,\mathbb{P}(A).$$

Conditioning on a die

Roll a fair die. What is $\mathbb{P}(\text{even} \mid \text{greater than } 3)$?

Solution. The conditioning event is $B = \{4, 5, 6\}$, so $\mathbb{P}(B) = 3/6$. The even outcomes inside B are $\{4, 6\}$, so $\mathbb{P}(A \cap B) = 2/6$. Hence $\mathbb{P}(A \mid B) = \frac{2/6}{3/6} = \frac{2}{3}$. Knowing the result exceeds 3 raised the chance of "even" from 1/2 to 2/3.

Law of total probability

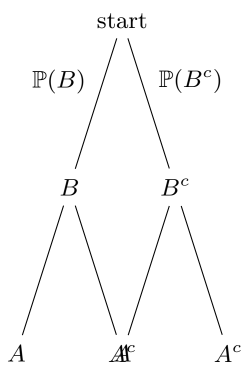
If the cases $B_1, B_2, \ldots, B_k$ are disjoint and together cover everything (a *partition*), then the probability of any event A is the weighted average of its conditional probabilities:

$$\mathbb{P}(A) = \sum_{i=1}^{k} \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i).$$

The two-case version, $\mathbb{P}(A) = \mathbb{P}(A \mid B)\mathbb{P}(B) + \mathbb{P}(A \mid B^c)\mathbb{P}(B^c)$, is the one we use for Bayes' theorem.

Intuition

To find how likely A is overall, split the world into scenarios, work out how likely A is *within* each scenario, and average — weighting each scenario by how likely it is. A probability tree makes this automatic: multiply along branches, then add across the branches that end in A.



$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(B)\,\mathbb{P}(A|B) \\ &\quad + \mathbb{P}(B^c)\,\mathbb{P}(A|B^c)\end{aligned}$$

Two boxes

Box 1 holds 3 red and 1 blue token; Box 2 holds 1 red and 3 blue. You pick a box at random (each with probability 1/2) and draw one token. What is $\mathbb{P}(\text{red})$?

Solution. By the law of total probability,

$$\mathbb{P}(\text{red}) = \mathbb{P}(\text{red} \mid B_1)\mathbb{P}(B_1) + \mathbb{P}(\text{red} \mid B_2)\mathbb{P}(B_2) = \tfrac{3}{4} \cdot \tfrac{1}{2} + \tfrac{1}{4} \cdot \tfrac{1}{2} = \tfrac{3}{8} + \tfrac{1}{8} = \tfrac{1}{2}.$$

Lecturer note: Next we will reverse *this: given a red token, which box was it probably from? That is Bayes, and the $\mathbb{P}(\text{red}) = \frac{1}{2}$ here is exactly its denominator.*

The multiplication rule also chains to more than two events (the **chain rule**), which we will need for sequences of trials:

$$\mathbb{P}(A_1 \cap A_2 \cap A_3) = \mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1)\,\mathbb{P}(A_3 \mid A_1 \cap A_2).$$

Definition

Events A and B are **independent** if knowing one tells us nothing about the other:

$$\mathbb{P}(A \mid B) = \mathbb{P}(A), \quad \text{equivalently} \quad \mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B).$$

Common pitfall

Independent and *disjoint* are opposites, not synonyms. Disjoint events cannot co-occur, so learning that one happened tells you the other did *not* — that is maximal *dependence*. Independence means the events carry no information about each other. Students conflate these constantly.

Why it matters for cognitive science

Independence assumptions are what make models computable. The "naive Bayes" classifier assumes features are conditionally independent given the category; many models of memory and categorisation assume trials are independent. These assumptions are rarely exactly true, but they are often good enough — and knowing *when* they break is a core modeling skill.

1.5 Bayes' theorem — updating belief (≈*35 min*)

Key idea

If students remember one formula from the whole course, it is this one. Bayes' theorem tells us how to *flip* a conditional probability and, in doing so, how to update a belief in light of evidence:

$$\boxed{\mathbb{P}(H \mid E) = \frac{\mathbb{P}(E \mid H)\,\mathbb{P}(H)}{\mathbb{P}(E)}}$$

where H is a *hypothesis* and E is the *evidence* we observe.

The four pieces have names worth drilling:

- $\mathbb{P}(H)$ — the **prior**: our belief before seeing the evidence.
- $\mathbb{P}(E \mid H)$ — the **likelihood**: how well the hypothesis predicts the evidence.
- $\mathbb{P}(H \mid E)$ — the **posterior**: our updated belief.
- $\mathbb{P}(E)$ — the **evidence** (or marginal likelihood): a normalising constant.

The denominator is almost always computed with the **law of total probability**: if the hypotheses H and H^c cover all cases,

$$\mathbb{P}(E) = \mathbb{P}(E \mid H)\,\mathbb{P}(H) + \mathbb{P}(E \mid H^c)\,\mathbb{P}(H^c).$$

Intuition

Read Bayes as a sentence: *new belief* $\propto$ *how well the hypothesis predicted what I saw* $\times$ *how much I believed it beforehand.* A hypothesis wins support when it both predicted the evidence well *and* was plausible to begin with. The denominator just rescales the competing hypotheses so they sum to one again.

The classic medical test

A disease affects 1% of a population. A test correctly flags 99% of true cases (sensitivity) and wrongly flags 5% of healthy people (false-positive rate). A random person tests positive. What is the probability they have the disease?

Solution. Let D = has disease, $+$ = positive test. We want $\mathbb{P}(D \mid +)$. Priors and likelihoods:

$$\mathbb{P}(D) = 0.01, \quad \mathbb{P}(D^c) = 0.99, \quad \mathbb{P}(+ \mid D) = 0.99, \quad \mathbb{P}(+ \mid D^c) = 0.05.$$

Total probability of a positive test:

$$\mathbb{P}(+) = \underbrace{0.99 \times 0.01}_{=0.0099} + \underbrace{0.05 \times 0.99}_{=0.0495} = 0.0594.$$

Bayes:

$$\mathbb{P}(D \mid +) = \frac{0.99 \times 0.01}{0.0594} = \frac{0.0099}{0.0594} = \frac{1}{6} \approx 16.7\%.$$

Despite a very accurate test, a positive result means only a $\sim$17% chance of disease — because the disease is rare, most positives come from the large healthy group.

Lecturer note: Draw this as 10,000 people: 100 sick (99 test positive), 9,900 healthy (495 test positive). Then $99/(99+495) = 1/6$ falls out with no algebra. Natural frequencies make the base-rate effect obvious.

Why it matters for cognitive science

This example *is* the **base-rate fallacy**: people (including doctors) systematically overestimate $\mathbb{P}(D \mid +)$ because they neglect the low prior $\mathbb{P}(D)$. It is one of the most robust findings in the psychology of judgment (Kahneman & Tversky) and a staple of any cognitive science curriculum. The same mathematics underlies Bayesian models of perception, where the "prior" is the brain's expectation and the "likelihood" is the sensory evidence.

Reversing the two boxes

Using the two-box setup, you drew a *red* token. What is the probability it came from Box 1?

Solution. We want $\mathbb{P}(B_1 \mid \text{red})$. We already found the denominator $\mathbb{P}(\text{red}) = \frac{1}{2}$. Then

$$\mathbb{P}(B_1 \mid \text{red}) = \frac{\mathbb{P}(\text{red} \mid B_1)\mathbb{P}(B_1)}{\mathbb{P}(\text{red})} = \frac{\frac{3}{4} \cdot \frac{1}{2}}{\frac{1}{2}} = \frac{3}{4}.$$

Seeing red raised our belief in Box 1 from $\frac{1}{2}$ (prior) to $\frac{3}{4}$ (posterior), because Box 1 predicts red better.

The odds form of Bayes — often the fastest

Write the *prior odds* of H versus its alternative, multiply by the *likelihood ratio*, and read off the *posterior odds*:

$$\underbrace{\frac{\mathbb{P}(H \mid E)}{\mathbb{P}(H^c \mid E)}}_{\text{posterior odds}} = \underbrace{\frac{\mathbb{P}(E \mid H)}{\mathbb{P}(E \mid H^c)}}_{\text{likelihood ratio}} \times \underbrace{\frac{\mathbb{P}(H)}{\mathbb{P}(H^c)}}_{\text{prior odds}}.$$

The normalising constant $\mathbb{P}(E)$ cancels, so this form needs no denominator.

Intuition

Evidence *multiplies* odds. A likelihood ratio of 10 means the evidence is ten times more expected under H than under its alternative, so it multiplies your odds tenfold — whatever they were. This is exactly how a diagnostic test, or a piece of behavioural evidence, "moves the needle". Re-doing the medical test in odds: prior odds 1:99, likelihood ratio $0.99/0.05 \approx 19.8$, so posterior odds $\approx 19.8{:}99 \approx 1{:}5$ — the same $1/6$ as before.

Bayes — do together

A jar of candies is either *ordinary* (30% red) or *special* (70% red); a priori each is equally likely. You draw one candy and it is red.

1. What is $\mathbb{P}(\text{red})$?

   Answer. $0.7 \cdot 0.5 + 0.3 \cdot 0.5 = 0.5$.
2. What is $\mathbb{P}(\text{special} \mid \text{red})$?

   Answer. $\dfrac{0.7 \cdot 0.5}{0.5} = 0.7$.
3. Using odds: prior odds 1:1, likelihood ratio $0.7/0.3$, so posterior odds 7:3, i.e. probability 0.7.

   Answer. Consistent with (2).

1.6 Random variables (≈*25 min*)

So far outcomes have been labels (heads, even, positive). Usually we want *numbers* we can average and plot. A random variable makes that translation.

Definition

A **random variable** X is a rule that assigns a number to every outcome of an experiment. It is **discrete** if it takes isolated values $(0, 1, 2, \dots)$ and **continuous** if it can take any value in an interval.

Examples: the number of correct trials out of 20 (discrete); a reaction time in milliseconds (continuous); the count of spikes in a window (discrete); a voltage (continuous).

Key idea

Two functions describe where a random variable's probability "lives".

For a *discrete* X, the **probability mass function** $p(x) = \mathbb{P}(X = x)$ gives the probability of each value; the masses are non-negative and sum to 1.

For a *continuous* X, the **probability density function** $f(x)$ is such that areas under it give probabilities: $\mathbb{P}(a \le X \le b) = \int_a^b f(x)\,dx$, and the total area is 1.

For *either*, the **cumulative distribution function** $F(x) = \mathbb{P}(X \le x)$ accumulates probability from the left.

Common pitfall

For a continuous variable, $f(x)$ is a *density*, not a probability: it can exceed 1, and $\mathbb{P}(X = x) = 0$ for any single point. Only *areas* (intervals) carry probability. This trips up every beginner — stress that "the probability of exactly $300.000\dots$ ms" is zero, but "between 290 and 310 ms" is not.

Intuition

A PMF is a bar chart of probabilities; a PDF is the smooth curve you get when the bars become infinitely thin. The CDF is the same information told as "how much probability have we accumulated so far", rising from 0 to 1.

What every PMF/PDF and CDF must satisfy

PMF: $p(x) \ge 0$ and $\sum_x p(x) = 1$. **PDF:** $f(x) \ge 0$ and $\int_{-\infty}^{\infty} f(x)\,dx = 1$.
CDF: F is non-decreasing, $F(-\infty) = 0$, $F(+\infty) = 1$, and $\mathbb{P}(a < X \le b) = F(b) - F(a)$.

Finding a missing probability

A discrete response variable $X \in \{0, 1, 2, 3\}$ has $p(0) = 0.1$, $p(1) = 0.3$, $p(2) = 0.4$, and $p(3)$ unknown. Find $p(3)$ and $\mathbb{P}(X \ge 2)$.

Solution. The masses must sum to 1, so $p(3) = 1 - (0.1 + 0.3 + 0.4) = 0.2$. Then $\mathbb{P}(X \ge 2) = p(2) + p(3) = 0.4 + 0.2 = 0.6$.

Random variables — do together

A continuous variable has density $f(x) = 2x$ for $0 \le x \le 1$ and 0 elsewhere.

1. Check it is a valid density.

   Answer. $\int_0^1 2x\,dx = [x^2]_0^1 = 1$, and $f \ge 0$. Valid.

2. Find $\mathbb{P}(X \le 0.5)$.

Answer. $\int_0^{0.5} 2x\,dx = [x^2]_0^{0.5} = 0.25$.

3. Why is $\mathbb{P}(X = 0.5) = 0$?

   Answer. A single point has zero width, so zero area under the density.

1.7 The essential distributions (≈*30 min*)

A distribution is a named, reusable shape of randomness. Four cover most of what students will meet. We give each one its story, not just its formula.

1.7.1 Bernoulli — a single yes/no trial

A **Bernoulli** variable is 1 with probability p ("success") and 0 with probability $1 - p$. It is the atom of all binary data: one coin flip, one correct/incorrect trial, one detected/undetected signal.

$$p(1) = p, \qquad p(0) = 1 - p, \qquad \mathbb{E}[X] = p, \qquad \mathrm{Var}(X) = p(1 - p).$$

Where $\mathbb{E}[X] = p$ and $\mathrm{Var}(X) = p(1 - p)$ come from

Directly from the definitions (which we formalise in Session 2): $\mathbb{E}[X] = 1 \cdot p + 0 \cdot (1 - p) = p$. Since $X^2 = X$ for a 0/1 variable, $\mathbb{E}[X^2] = p$, so $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = p - p^2 = p(1 - p)$. The variance is largest at $p = \frac{1}{2}$ (a fair coin is the most unpredictable) and zero at $p = 0$ or 1.

1.7.2 Binomial — counting successes in n trials

If we repeat n *independent* Bernoulli trials each with success probability p, the number of successes X is **Binomial**(n, p):

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \qquad \mathbb{E}[X] = np, \qquad \mathrm{Var}(X) = np(1 - p).$$

Guessing on a multiple-choice test

A student guesses on 10 four-option questions ($p = 1/4$). What is the probability of exactly 3 correct? How many do we expect?

Solution. With $n = 10$, $p = 0.25$, $k = 3$:

$$\mathbb{P}(X = 3) = \binom{10}{3} (0.25)^3 (0.75)^7 = 120 \times 0.015625 \times 0.13349 \approx 0.250.$$

Expected number correct: $\mathbb{E}[X] = np = 10 \times 0.25 = 2.5$.

1.7.3 Poisson — counts of rare events

When events happen independently at a constant average rate, the number in a fixed window is **Poisson**(λ), where λ is both the mean and the variance:

$$\mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \qquad \mathbb{E}[X] = \lambda, \qquad \mathrm{Var}(X) = \lambda.$$

Why it matters for cognitive science

The Poisson distribution is the standard first model of **neural spike counts**: if a neuron fires at an average rate λ in a time window, the observed count is roughly Poisson. The tell-tale signature "variance = mean" is used constantly to check whether real spike data are more or less variable than this baseline (over- or under-dispersion).

Spike counts

A neuron fires on average $\lambda = 5$ spikes per second. What is the probability of observing exactly 3 spikes in a given second?

Solution.

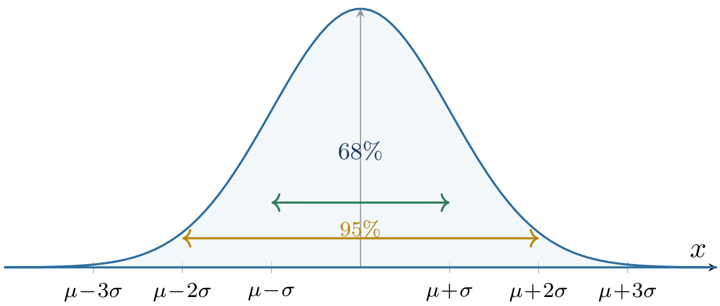
$$\mathbb{P}(X = 3) = \frac{5^3 e^{-5}}{3!} = \frac{125 \times 0.006738}{6} \approx 0.140.$$

1.7.4 Gaussian — the bell curve

The **Normal** (Gaussian) distribution $\mathcal{N}(\mu, \sigma^2)$ is the continuous workhorse, with mean μ and standard deviation σ:

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

It is symmetric and bell-shaped, centred at μ, with spread set by σ.



Key idea

The 68–95–99.7 rule. For any Normal distribution, about 68% of the probability lies within $\pm 1\sigma$ of the mean, about 95% within $\pm 2\sigma$, and about 99.7% within $\pm 3\sigma$. This single fact lets students reason about Normal data without any integration.

To compute Normal probabilities we **standardise**: convert to the number of standard deviations from the mean, $z = \dfrac{x-\mu}{\sigma}$, which follows the *standard* Normal $\mathcal{N}(0, 1)$.

Standardising

Adult heights in a population are approximately $\mathcal{N}(\mu = 170, \sigma = 10)$ cm. What fraction of people are taller than 185 cm?

Solution. Standardise: $z = \dfrac{185 - 170}{10} = 1.5$. From a z-table (or the rule), $\mathbb{P}(Z > 1.5) \approx 0.067$, so about 6.7% of people exceed 185 cm.

1.7.5 Two more worth naming

The **Uniform** distribution spreads probability evenly over an interval ("no value preferred"); it is the natural model of complete ignorance and the basis of random number generation. The

Exponential distribution models *waiting times* between Poisson events (e.g. inter-spike intervals) and is the continuous cousin of the Poisson counts.

Why it matters for cognitive science

Why does the Gaussian appear everywhere — test scores, measurement error, averaged neural signals? Because of the *Central Limit Theorem*, which we prove intuitively next session: whenever a quantity is the sum of many small, independent influences, its distribution is approximately Normal regardless of the details. This is why Gaussian noise is the default assumption in almost every model of perception and data analysis.

The four distributions at a glance

Distribution	Models	Support	Mean	Variance
Bernoulli(p)	one yes/no trial	$\{0, 1\}$	p	$p(1-p)$
Binomial(n, p)	# successes in n trials	$\{0, \ldots, n\}$	np	$np(1-p)$
Poisson(λ)	counts of rare events	$\{0, 1, 2, \ldots\}$	λ	λ
Normal(μ, σ^2)	bell-shaped noise/averages	$\mathbb{R}$	μ	σ^2

Two connections: a Binomial is a *sum* of n independent Bernoullis; and a Binomial with large n is well approximated by a Normal (a special case of the CLT), or by a Poisson when n is large and p small.

Distributions — do together

1. A subject presses a button correctly on 70% of trials, independently. In $n = 8$ trials, what is the expected number correct, and $\mathbb{P}(\text{all 8 correct})$?

   Answer. $\mathbb{E} = np = 5.6$; $\mathbb{P}(8) = 0.7^8 \approx 0.0576$.

2. A neuron fires at $\lambda = 2$ spikes per window. What is $\mathbb{P}(X = 0)$ (no spike)?

   Answer. $e^{-2} \approx 0.135$.

3. Reaction times are $\mathcal{N}(\mu = 500, \sigma = 50)$ ms. Roughly what fraction fall between 450 and 550 ms?

   Answer. within $\pm 1\sigma$, so about 68%.

Session recap

Today we built the language of uncertainty. Probabilities are numbers in $[0, 1]$ obeying the complement and addition rules. *Conditional probability* $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$ formalises learning from information, and *independence* is its absence. *Bayes' theorem*, $\mathbb{P}(H \mid E) \propto \mathbb{P}(E \mid H)\mathbb{P}(H)$, is how evidence updates belief — the mathematical core of Bayesian cognitive science and the base-rate fallacy. Finally, *random variables* turn outcomes into numbers, described by a PMF/PDF and CDF, and four *distributions* — Bernoulli, Binomial, Poisson, Gaussian — model binary trials, counts of successes, counts of rare events, and bell-shaped noise respectively. Next session we turn from single random quantities to *data*: summarising it, and inferring from it.

Chapter 2

Probability & Statistics II — From Data to Inference

Where this session is going

Session 1 described *single* uncertain quantities. Today we turn to *data*: many observations at once. We learn to summarise a distribution with a few numbers (*mean, variance, covariance*), to understand why averages are trustworthy (*Law of Large Numbers* and *Central Limit Theorem*), and then to *infer* from a sample to a population — confidence intervals, hypothesis tests, and the first genuine model of the course, *linear regression*. The thread running through everything is the distinction between what is true in the *population* and what we happen to see in a *sample*.

2.1 Summarising a distribution (≈*30 min*)

A distribution can be complicated, but a few numbers capture most of what we care about: where it is centred, and how spread out it is.

2.1.1 Expectation — the centre of mass

Definition

The **expectation** (mean) of a random variable is its probability-weighted average value:

$$\mathbb{E}[X] = \sum_x x\,p(x) \quad \text{(discrete)}, \qquad \mathbb{E}[X] = \int x\,f(x)\,dx \quad \text{(continuous)}.$$

Intuition

Picture the distribution as a pile of mass on the number line. The expectation is its *balance point* — where you would place a finger to keep it level. It need not be a value the variable can actually take (a die has mean 3.5).

The property that makes expectation so useful is **linearity**, which holds *always*, even when variables are dependent:

$$\mathbb{E}[aX + bY + c] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y] + c.$$

Expectation of a die, and of a shifted score

A fair die X has $\mathbb{E}[X] = \frac{1}{6}(1+2+3+4+5+6) = \frac{21}{6} = 3.5$. If a game pays $Y = 2X+1$ points, then by linearity $\mathbb{E}[Y] = 2 \cdot 3.5 + 1 = 8$, with no need to re-average the six payouts.

2.1.2 Variance and standard deviation — the spread

Definition

The **variance** measures the average squared distance from the mean, and the **standard deviation** is its square root (back in the original units):

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big], \qquad \mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

Intuition

Variance answers "how far, on average, do outcomes fall from the centre?" We square the deviations so that positive and negative ones do not cancel, and to punish large deviations more. The standard deviation undoes the squaring so the spread is expressed in the same units as the data (milliseconds, not milliseconds-squared).

Common pitfall

Variance does *not* share expectation's simple linearity. In general $\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$ (note the square), and $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ only when X and Y are *independent*. The correction term for the dependent case is exactly the covariance, defined next.

Variance rules worth memorising

- **Computational formula:** $\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2$ ("mean of the square minus square of the mean") — usually the fastest way to compute by hand.
- **Shift and scale:** $\mathrm{Var}(aX+b) = a^2\ \mathrm{Var}(X)$; adding a constant does not change spread, and scaling multiplies the *standard deviation* by $|a|$.
- **Sum:** $\mathrm{Var}(X+Y) = \mathrm{Var}(X)+\mathrm{Var}(Y)+2\ \mathrm{Cov}(X,Y)$, which reduces to $\mathrm{Var}(X)+\mathrm{Var}(Y)$ exactly when X and Y are uncorrelated.

Variance of a die

Find $\mathrm{Var}(X)$ for a fair die.

Solution. Use the computational formula. We have $\mathbb{E}[X] = 3.5$ and $\mathbb{E}[X^2] = \frac{1}{6}(1+4+9+16+25+36) = \frac{91}{6} \approx 15.17$. Hence $\mathrm{Var}(X) = \frac{91}{6} - 3.5^2 = 15.17 - 12.25 = 2.9167$, so $\mathrm{SD}(X) = \sqrt{2.9167} \approx 1.71$.

2.1.3 Covariance and correlation — how two variables move together

Definition

The **covariance** of X and Y measures whether they tend to be large or small together:

$$\mathrm{Cov}(X,Y) = \mathbb{E}\big[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\big].$$

The **correlation** rescales it to lie in $[-1,1]$:

$$\mathrm{Corr}(X,Y)=\rho=\frac{\mathrm{Cov}(X,Y)}{\mathrm{SD}(X)\ \mathrm{SD}(Y)}.$$

Intuition

Covariance is positive when the two variables are usually on the same side of their means (both above, or both below), and negative when one is high while the other is low. But its size depends on the units, so it is hard to read directly. Correlation fixes this: $\rho=+1$ is a perfect increasing line, $\rho=-1$ a perfect decreasing line, $\rho=0$ no *linear* relationship.

Key idea

Covariance is the bridge between statistics and linear algebra. When we have many variables at once, all their pairwise covariances are collected into a **covariance matrix** — the central object of Session 4's Principal Component Analysis. Keep this word in mind; it returns.

Covariance and correlation by hand

Five students' weekly study hours x and quiz scores y:

$$x=(1,2,3,4,5),\qquad y=(2,4,5,4,5).$$

Solution. The means are $\bar{x}=3$ and $\bar{y}=4$. Tabulate the deviations:

						sum
$x-\bar{x}$	-2	-1	0	1	2	0
$y-\bar{y}$	-2	0	1	0	1	0
$(x-\bar{x})(y-\bar{y})$	4	0	0	0	2	6
$(x-\bar{x})^2$	4	1	0	1	4	10
$(y-\bar{y})^2$	4	0	1	0	1	6

The correlation does not depend on whether we divide by n or $n-1$, because the factor cancels:

$$\rho=\frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2\,\sum(y-\bar{y})^2}}=\frac{6}{\sqrt{10\times 6}}=\frac{6}{\sqrt{60}}\approx 0.77.$$

A strong positive linear association: more study hours go with higher scores.

Lecturer note: We reuse the sums 6, 10, 6 for the regression line at the end of the session — point this out so students see the two topics are one.

Covariance facts

- $\mathrm{Cov}(X,X)=\mathrm{Var}(X)$: variance is "self-covariance".
- Symmetry: $\mathrm{Cov}(X,Y)=\mathrm{Cov}(Y,X)$; and shifting does not matter, $\mathrm{Cov}(X+a,\ Y+b)=\mathrm{Cov}(X,Y)$.
- Scaling: $\mathrm{Cov}(aX,\ bY)=ab\ \mathrm{Cov}(X,Y)$, so correlation (which divides by the two SDs) is *unit-free* and lies in $[-1,1]$.
- Independence $\Rightarrow$ Cov $=0$, but **not** the reverse: zero covariance means no *linear* link, not independence.

Summaries — do together

A variable takes values $\{2, 4, 6\}$ each with probability $1/3$.

1. Find $\mathbb{E}[X]$.

   Answer. $(2 + 4 + 6)/3 = 4$.
2. Find $\mathrm{Var}(X)$ with the computational formula.

   Answer. $\mathbb{E}[X^2] = (4 + 16 + 36)/3 = \frac{56}{3} \approx 18.67$; $\mathrm{Var} = 18.67 - 16 = 2.67$.
3. If $Y = 3X$, what is $\mathrm{SD}(Y)$?

   Answer. $\mathrm{SD}(X) = \sqrt{2.67} \approx 1.63$, so $\mathrm{SD}(Y) = 3 \cdot 1.63 \approx 4.90$.

Common pitfall

$\rho = 0$ means no *linear* relationship, not "no relationship". Points on a perfect parabola $y = x^2$ (symmetric about 0) have zero correlation despite being perfectly predictable. And correlation is never, by itself, evidence of causation — a lesson worth repeating in a cognitive-science course full of observational data.

2.2 Why averages are trustworthy (≈*25 min*)

Two theorems justify the entire practice of estimating things by averaging.

Key idea

Law of Large Numbers (LLN). As the sample size n grows, the sample mean $\bar{X}_n$ converges to the true mean μ. *Averages settle down.*

Central Limit Theorem (CLT). For large n, the sample mean is approximately Normal, $\bar{X}_n \approx \mathcal{N}(\mu,\ \sigma^2/n)$, *whatever the shape* of the original distribution. *Sums and averages become bell-shaped.*

Intuition

The LLN says the average of many trials lands near the truth — it is why more data is better. The CLT says something stronger and more surprising: it tells us the *shape* of the leftover error. Even if individual observations are wildly non-Normal (a skewed reaction-time, a 0/1 correct response), their average over many trials traces out a Normal curve, and its spread shrinks like $\sigma/\sqrt{n}$. This is why the Normal distribution is unavoidable, and why error bars have the form they do.

Why it matters for cognitive science

The CLT explains why so many measured quantities in psychology and neuroscience look Gaussian: each is the aggregate of many small independent contributions (many neurons, many trials, many genes and experiences behind a trait). It also underwrites the ubiquitous practice of trial-averaging in EEG/fMRI: averaging n noisy trials cuts the noise standard deviation by $\sqrt{n}$, sharpening a buried signal.

The $\sqrt{n}$ law of averaging

If $X_1, \ldots, X_n$ are independent with the same mean μ and SD σ, then the sample mean $\bar{X}$ has

$$\mathbb{E}[\bar{X}] = \mu \quad \text{(centred correctly)}, \qquad \mathrm{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}} \quad \text{(spread shrinks)}.$$

To *halve* the uncertainty of an estimate you need *four times* the data; to cut it tenfold, a hundredfold. This diminishing return is why large-n studies are expensive.

How many trials to halve the noise?

Single-trial EEG responses have noise SD $\sigma = 10\,\mu$V. You currently average 25 trials. How many trials would halve the noise of the average?

Solution. With 25 trials the averaged noise is $10/\sqrt{25} = 2\,\mu$V. To reach $1\,\mu$V we need $10/\sqrt{n} = 1$, i.e. $n = 100$: four times as many trials, as the $\sqrt{n}$ law predicts.

2.3 From sample to population: estimation (≈*25 min*)

We almost never see a whole population; we see a *sample* and use it to guess population quantities.

Definition

A **parameter** is a fixed but unknown number describing the population (e.g. the true mean μ). An **estimator** is a recipe applied to the sample to approximate it (e.g. the sample mean $\bar{X}$). An estimator is **unbiased** if it is correct *on average* across many samples: $\mathbb{E}[\hat{\theta}] = \theta$.

Common pitfall

When estimating variance from a sample we divide by $n-1$, not n:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Using the sample mean $\bar{x}$ (rather than the unknown true mean) makes the deviations slightly too small; dividing by $n-1$ instead of n corrects the bias. Students should know the rule and its reason, not just the formula.

The key quantity for inference is how much an estimate would *wobble* from sample to sample.

Definition

The **standard error** of the mean is the standard deviation of the sample mean across hypothetical repeated samples:

$$\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}} \quad \text{(estimated by } s/\sqrt{n}\text{)}.$$

Common pitfall

Standard deviation and standard error are different. The *standard deviation* s describes the spread of the raw data and does not shrink with more data. The *standard error* $s/\sqrt{n}$ describes the precision of our *estimate of the mean* and does shrink as n grows. Reporting one when you mean the other is one of the most common errors in published figures.

When σ is unknown: use t instead of z

In practice we rarely know the population SD σ; we estimate it by the sample SD s. This extra uncertainty means the standardised mean follows *Student's t-distribution* with $n-1$ degrees of freedom rather than the Normal. The t-curve looks Normal but with heavier tails;

for small samples the 95% multiplier is larger than 1.96 (e.g. about 2.26 for $n = 10$), and it approaches 1.96 as n grows. Rule of thumb: for $n \gtrsim 30$ the difference is negligible.

2.4 Confidence intervals (≈*20 min*)

A single number hides its own uncertainty. A confidence interval reports a *range* of plausible values for the parameter.

Definition

A **95% confidence interval** for the mean is

$$\bar{x} \pm 1.96\,\frac{\sigma}{\sqrt{n}}.$$

The recipe is constructed so that, across many repeated samples, 95% of the intervals it produces contain the true μ.

Intuition

The correct reading is about the *procedure*, not a single interval: if we repeated the experiment many times, 95% of the intervals we build this way would capture the truth. The width $\pm 1.96\,\sigma/\sqrt{n}$ comes straight from the CLT and the 95% mark of the Normal curve.

A confidence interval

A test with known population SD $\sigma = 15$ is given to $n = 25$ people; the sample mean is $\bar{x} = 105$. Give a 95% confidence interval for the population mean.

Solution. Standard error: $\mathrm{SE} = 15/\sqrt{25} = 15/5 = 3$. Interval: $105 \pm 1.96 \times 3 = 105 \pm 5.88$, i.e. about $[99.1,\ 110.9]$.

Common pitfall

"95% confidence" does *not* mean "there is a 95% probability the true mean is in this particular interval". In the classical framework μ is fixed; it is the interval that is random. Getting this subtlety right is exactly the kind of care that distinguishes a competent data analyst.

2.5 Hypothesis testing (≈*30 min*)

The dominant — and most criticised — ritual of statistics. Students must be able to both perform it and understand its limits.

Key idea

The logic in four steps.

1. State a **null hypothesis** H_0 ("nothing is going on") and an **alternative** H_1.
2. Choose a **test statistic** that would be extreme if H_1 were true.
3. Compute the p**-value**: the probability, *assuming* H_0, of a result at least as extreme as the one observed.
4. If p is below a threshold α (often 0.05), **reject** H_0; otherwise fail to reject it.

Is the coin fair?

A coin lands heads 60 times in 100 flips. Test $H_0 : p = 0.5$ against $H_1 : p \neq 0.5$.

Solution. Under H_0, the number of heads has mean $np = 50$ and standard deviation $\sqrt{np(1-p)} = \sqrt{100 \cdot 0.25} = 5$. The standardised statistic is

$$z = \frac{60 - 50}{5} = 2.0.$$

A z of 2.0 or more extreme (either tail) has probability $\mathbb{P}(|Z| \geq 2.0) \approx 0.046$. Since $0.046 < 0.05$, we reject H_0 at the 5% level: the coin shows weak but statistically significant evidence of bias.

Definition

Two ways a test can be wrong: *Type I error* — rejecting a true H_0 (a false positive), with probability α; *Type II error* — failing to reject a false H_0 (a false negative), with probability β. The **power** of a test is $1 - \beta$: its chance of detecting a real effect.

The workhorse tests, all following the same logic, differ only in the statistic: the ***t*-test** compares means (one sample, or two groups); the χ^2 **test** compares observed vs. expected counts in categories; the **ANOVA** (F-test) compares means across three or more groups. Each carries assumptions (roughly: independence, and approximate Normality of the relevant averages) worth stating explicitly.

A two-group comparison

Group A ($n_A = 25$) scores a mean of 105; group B ($n_B = 25$) a mean of 100; the pooled standard deviation is $s = 10$. Is the difference significant, and how large is it?

Solution. The standard error of the *difference* in means is

$$\text{SE} = s\sqrt{\tfrac{1}{n_A} + \tfrac{1}{n_B}} = 10\sqrt{\tfrac{1}{25} + \tfrac{1}{25}} = 10\sqrt{0.08} \approx 2.83.$$

The t-statistic is $t = \dfrac{105 - 100}{2.83} \approx 1.77$ on about 48 degrees of freedom, giving a two-sided $p \approx 0.08$ — *not* significant at 0.05. Yet the **effect size** is $d = \dfrac{105 - 100}{10} = 0.5$, a "medium" standardised difference. With only 25 per group the study is simply underpowered to detect it.

Lecturer note: This is the ideal example for separating "significant" from "big": a real, medium effect that the test misses for lack of power.

Why it matters for cognitive science

The p-value is routinely misunderstood, including by researchers. It is *not* the probability that H_0 is true, nor the probability the result was "due to chance". Misuse of $p < 0.05$ — through multiple comparisons, optional stopping, and publication bias — is a central driver of the **replication crisis** that has reshaped psychology and cognitive neuroscience since the 2010s. Teaching students to report *effect sizes* and *confidence intervals* alongside (or instead of) p-values is part of the modern remedy.

Common pitfall

“Statistically significant” is not the same as “important”. With a large enough n, a trivially small effect can reach $p < 0.05$; with a small n, a real and large effect can be missed. Always ask “how big is the effect?” (an effect size such as Cohen’s d or the correlation ρ), not only “is $p < 0.05$?”.

Inference — do together

1. A sample of $n = 100$ has mean $\bar{x} = 52$ with known $\sigma = 10$. Give a 95% CI for μ.
   Answer. $\mathrm{SE} = 10/\sqrt{100} = 1$; $\mathrm{CI} = 52 \pm 1.96$, i.e. $[50.04,\ 53.96]$.
2. Testing $H_0 : \mu = 50$ against $\mu \neq 50$, what is the z-statistic and rough conclusion?
   Answer. $z = (52 - 50)/1 = 2.0$, $p \approx 0.046$: reject at 5%.
3. A treatment raises scores by 2 points with within-group SD 10. What is Cohen’s d, and is it small, medium, or large?
   Answer. $d = 2/10 = 0.2$: a small effect.

2.6 The first model: linear regression (≈*25 min*)

Correlation measures the strength of a linear relationship; *regression* draws the actual line and lets us predict. This is the gateway to the whole of supervised machine learning.

Definition

Simple linear regression fits the line $\hat{y} = a + bx$ to data (x_i, y_i) by **least squares**: choosing the intercept a and slope b that minimise the sum of squared vertical errors (residuals) $\sum_i (y_i - \hat{y}_i)^2$. The solution is

$$b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\mathrm{Cov}(x, y)}{\mathrm{Var}(x)}, \qquad a = \bar{y} - b\,\bar{x}.$$

Why the least-squares line has this slope

Minimise $S(a, b) = \sum_i (y_i - a - bx_i)^2$. Setting the derivatives to zero gives the two “normal equations”. The one from $\partial S/\partial a = 0$ says the residuals sum to zero, forcing the line through the mean point $(\bar{x}, \bar{y})$, hence $a = \bar{y} - b\bar{x}$. Substituting and solving $\partial S/\partial b = 0$ gives $b = \sum (x_i - \bar{x})(y_i - \bar{y}) / \sum (x_i - \bar{x})^2$. **Consequences:** the residuals always sum to zero and are uncorrelated with x; and the line always passes through $(\bar{x}, \bar{y})$ — a useful sanity check.

Intuition

We are looking for the straight line that passes as close as possible to all the points. “As close as possible” is measured by the squared vertical gaps between each point and the line; least squares finds the line that makes their total as small as possible. The slope is just the covariance of x and y rescaled by the spread of x — the same covariance we met an hour ago.

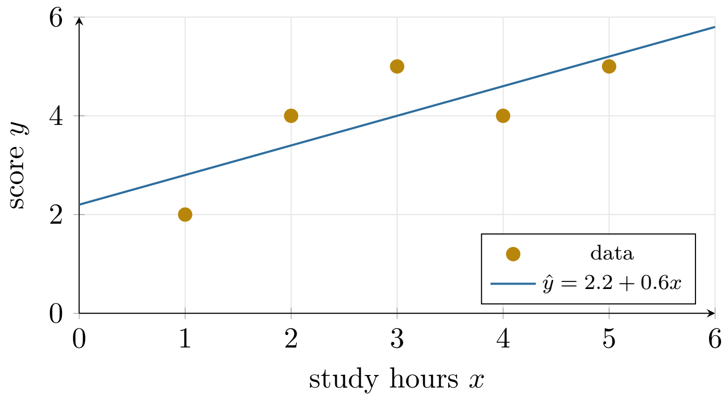
The regression line

Use the study-hours data $x = (1, 2, 3, 4, 5)$, $y = (2, 4, 5, 4, 5)$ from before.

Solution. We already computed $\sum(x-\bar{x})(y-\bar{y}) = 6$ and $\sum(x-\bar{x})^2 = 10$, with $\bar{x}=3,\ \bar{y}=4$. Hence

$$b = \frac{6}{10} = 0.6, \qquad a = 4 - 0.6\times 3 = 2.2, \qquad \Rightarrow \qquad \hat{y} = 2.2 + 0.6\,x.$$

Each extra study hour predicts 0.6 more points. The fraction of variance explained is $R^2 = \rho^2 = 0.77^2 \approx 0.60$.



Why it matters for cognitive science

Regression is the atom of statistical modeling. Extended to many predictors it becomes *multiple regression*; with a nonlinear "link" it becomes the *Generalized Linear Model* used for psychometric functions and neural encoding models (Session 5); and viewed as "fit parameters to minimise error" it is exactly the objective that supervised machine learning optimises. The idea of least squares also reappears geometrically in linear algebra as a *projection* (Session 3).

Regression — do together

Four points: $x = (0,1,2,3)$, $y = (1,1,2,4)$.

1. Find $\bar{x}, \bar{y}$.

   Answer. $\bar{x} = 1.5,\ \bar{y} = 2$.
2. Compute $\sum(x-\bar{x})(y-\bar{y})$ and $\sum(x-\bar{x})^2$.

   Answer. numerator $= (-1.5)(-1)+(-0.5)(-1)+(0.5)(0)+(1.5)(2) = 1.5+0.5+0+3 = 5$; denominator $= 2.25+0.25+0.25+2.25 = 5$.
3. Give the regression line.

   Answer. $b = 5/5 = 1$, $a = 2 - 1\cdot 1.5 = 0.5$, so $\hat{y} = 0.5 + x$.

Session recap

We moved from single variables to data and inference. A distribution is summarised by its *mean* (centre) and *variance/SD* (spread), and two variables by their *covariance/correlation*. The *LLN* guarantees averages converge to the truth and the *CLT* makes them Normal with spread $\sigma/\sqrt{n}$ — the engine behind *standard errors* and *confidence intervals*. *Hypothesis testing* formalises "is this more than chance?" via p-values, with the ever-present cautions about significance vs. importance and the replication crisis. Finally *linear regression* fits a predictive line by least squares, reusing covariance and foreshadowing both the GLM and machine learning. Next we switch subjects to *linear algebra* — but covariance, projection, and least squares will tie the two halves together.

Chapter 3

Linear Algebra I — Vectors, Matrices, and Geometry

Where this session is going

Linear algebra is the mathematics of *data in many dimensions* and of *linear transformations*. Today we build the objects and their geometry: a *vector* as a point/arrow/list of features; the *dot product* that measures similarity, length, and angle; the *matrix* read two ways — as a table of data and as a transformation of space; *matrix multiplication*; and finally *linear systems*, the *inverse*, and the ideas of *rank*, *independence*, and *basis* that tell us the true dimensionality of data. Next session these lead to eigenvectors and PCA.

3.1 Vectors: points, arrows, and features (≈*25 min*)

Definition

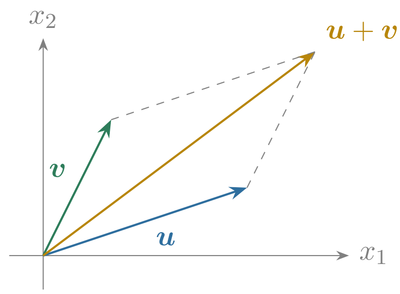
A **vector** is an ordered list of numbers, e.g. $\boldsymbol{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{R}^n$. We can read it in three equivalent ways: as a *point* at coordinates $(v_1, \ldots, v_n)$, as an *arrow* from the origin to that point, or as a *list of features* describing one object.

Intuition

The three readings are the whole game. "Arrow" gives us geometry (length, direction, angle). "Point" lets us picture a data set as a cloud of points. "List of features" is how real data arrives: a participant is a vector of (age, reaction time, accuracy, . . .); an image is a vector of pixel values; a word is a vector of context statistics. Linear algebra lets us treat all of these with one geometry.

Two operations define vectors. We add them *componentwise* (tip-to-tail geometrically) and we *scale* them by a number (stretching or flipping the arrow):

$$\boldsymbol{u} + \boldsymbol{v} = \left(\begin{smallmatrix} u_1+v_1 \\ u_2+v_2 \end{smallmatrix}\right), \qquad c\,\boldsymbol{v} = \left(\begin{smallmatrix} cv_1 \\ cv_2 \end{smallmatrix}\right).$$



Key idea

A **linear combination** $c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_k\boldsymbol{v}_k$ — scale some vectors and add them — is the single most important operation in the subject. Almost every concept to come (span, basis, matrix multiplication, regression, PCA) is a statement about linear combinations.

3.2 The dot product: similarity, length, angle (≈*30 min*)

One operation quietly encodes geometry: the dot product.

Definition

The **dot product** of two vectors is the sum of products of matching components:

$$\boldsymbol{u}\cdot\boldsymbol{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n = \sum_{i=1}^{n} u_iv_i.$$

It also satisfies the geometric identity $\boldsymbol{u}\cdot\boldsymbol{v} = \|\boldsymbol{u}\|\,\|\boldsymbol{v}\|\cos\theta$, where θ is the angle between the arrows.

From the dot product we get length and distance:

$$\|\boldsymbol{v}\| = \sqrt{\boldsymbol{v}\cdot\boldsymbol{v}} = \sqrt{v_1^2+\cdots+v_n^2} \quad \text{(the Euclidean/}L_2\text{ norm)}, \qquad \operatorname{dist}(\boldsymbol{u},\boldsymbol{v}) = \|\boldsymbol{u}-\boldsymbol{v}\|.$$

The L_1 norm $\|\boldsymbol{v}\|_1 = \sum_i |v_i|$ ("taxicab" distance) is a common alternative worth naming.

Intuition

The dot product is a *similarity score*. Vectors pointing the same way have a large positive dot product; perpendicular vectors have dot product zero ("nothing in common"); opposite vectors have a large negative one. Divide by the two lengths and you get $\cos\theta$ — the **cosine similarity** used everywhere to compare feature vectors, immune to their overall magnitude.

Length, dot product, angle

Let $\boldsymbol{u} = \left(\begin{smallmatrix}1\\2\end{smallmatrix}\right)$ and $\boldsymbol{v} = \left(\begin{smallmatrix}3\\1\end{smallmatrix}\right)$.

Solution. Dot product: $\boldsymbol{u}\cdot\boldsymbol{v} = 1\cdot 3 + 2\cdot 1 = 5$. Lengths: $\|\boldsymbol{u}\| = \sqrt{1+4} = \sqrt{5}$, $\|\boldsymbol{v}\| = \sqrt{9+1} = \sqrt{10}$. Angle:

$$\cos\theta = \frac{5}{\sqrt{5}\sqrt{10}} = \frac{5}{\sqrt{50}} = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}} \quad\Rightarrow\quad \theta = 45^\circ.$$

Key idea

Orthogonality. Two vectors are *perpendicular (orthogonal)* exactly when their dot product is zero. Orthogonality is the geometric meaning of "no linear relationship" — and it is the same idea as zero correlation once data are mean-centred. This connects directly to least-squares regression: the best-fit line is the one whose residual vector is orthogonal to the predictors.

Dot-product facts

- **Symmetry / linearity:** $\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{u}$ and $\boldsymbol{u} \cdot (\boldsymbol{v} + \boldsymbol{w}) = \boldsymbol{u} \cdot \boldsymbol{v} + \boldsymbol{u} \cdot \boldsymbol{w}$.
- **Cauchy–Schwarz inequality:** $|\boldsymbol{u} \cdot \boldsymbol{v}| \leq \|\boldsymbol{u}\| \, \|\boldsymbol{v}\|$, with equality only when the vectors are parallel. This is why cosine similarity always lies in $[-1, 1]$.
- **Triangle inequality:** $\|\boldsymbol{u} + \boldsymbol{v}\| \leq \|\boldsymbol{u}\| + \|\boldsymbol{v}\|$ (a detour is never shorter than the direct route).
- **Unit vector:** $\hat{\boldsymbol{u}} = \boldsymbol{u} / \|\boldsymbol{u}\|$ has length 1 and keeps only the direction.

Projection — the geometric core of least squares

The (orthogonal) projection of $\boldsymbol{b}$ onto the direction of $\boldsymbol{a}$ is

$$\operatorname{proj}_{\boldsymbol{a}} \boldsymbol{b} = \frac{\boldsymbol{a} \cdot \boldsymbol{b}}{\boldsymbol{a} \cdot \boldsymbol{a}} \, \boldsymbol{a}.$$

The leftover $\boldsymbol{b} - \operatorname{proj}_{\boldsymbol{a}} \boldsymbol{b}$ (the "residual") is orthogonal to $\boldsymbol{a}$: it is the part of $\boldsymbol{b}$ that $\boldsymbol{a}$ cannot explain.

Projecting one vector onto another

Project $\boldsymbol{b} = \binom{2}{3}$ onto $\boldsymbol{a} = \binom{1}{1}$.

Solution. $\boldsymbol{a} \cdot \boldsymbol{b} = 2 + 3 = 5$ and $\boldsymbol{a} \cdot \boldsymbol{a} = 2$, so $\operatorname{proj}_{\boldsymbol{a}} \boldsymbol{b} = \frac{5}{2}\binom{1}{1} = \binom{2.5}{2.5}$. The residual is $\binom{-0.5}{0.5}$, and indeed $\boldsymbol{a} \cdot \binom{-0.5}{0.5} = 0$: perpendicular, as it must be.

Lecturer note: Say explicitly: fitting a regression line is projecting the data vector onto the space spanned by the predictors. Session 2 and today are one idea.

Vectors & dot products — do together

Let $\boldsymbol{u} = \binom{3}{4}$, $\boldsymbol{v} = \binom{4}{-3}$.

1. Find $\|\boldsymbol{u}\|$.

   Answer. $\sqrt{9 + 16} = 5$.
2. Compute $\boldsymbol{u} \cdot \boldsymbol{v}$; are they orthogonal?

   Answer. $12 - 12 = 0$: yes, perpendicular.
3. Give the unit vector in the direction of $\boldsymbol{u}$.

   Answer. $\frac{1}{5}\binom{3}{4} = \binom{0.6}{0.8}$.

Why it matters for cognitive science

Representing stimuli, words, or neural population states as vectors and comparing them by dot product / cosine similarity is the backbone of *representational similarity analysis* in neuroscience and of *word embeddings* in computational linguistics. "King − man + woman ≈ queen" is literally vector arithmetic. A population of neurons firing is a point in a high-dimensional space, and how two conditions differ is a distance between two such points.

3.3 Matrices: data tables and transformations (≈*35 min*)

Definition

A **matrix** is a rectangular array of numbers with m rows and n columns ($m \times n$). It has *two readings*: a **data table** (each row an observation, each column a feature), and a **linear transformation** (a rule $\boldsymbol{x} \mapsto \boldsymbol{A}\boldsymbol{x}$ that moves every vector to a new place).

Key idea

Teach both readings and keep switching between them — this dual view is the key insight of the whole subject. The data-table reading is how we *store* observations; the transformation reading is how models *act* on them (a layer of a neural network multiplies its input by a weight matrix).

3.3.1 Matrix–vector multiplication

The product $\boldsymbol{A}\boldsymbol{x}$ is a *linear combination of the columns of* $\boldsymbol{A}$, weighted by the entries of $\boldsymbol{x}$. Equivalently, each entry of the output is a dot product of a row of $\boldsymbol{A}$ with $\boldsymbol{x}$:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax+by \\ cx+dy \end{pmatrix} = x\begin{pmatrix} a \\ c \end{pmatrix} + y\begin{pmatrix} b \\ d \end{pmatrix}.$$

Intuition

A matrix *transforms space*: it sends the grid of all vectors to a new grid, keeping the origin fixed and straight lines straight. The columns of $\boldsymbol{A}$ tell you exactly where the basis vectors land — the first column is where $\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$ goes, the second where $\left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)$ goes. Everything else follows by linear combination. Rotations, reflections, scalings, and shears are all just different matrices.

A matrix that rotates

The matrix $\boldsymbol{R} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ rotates the plane by 90° counter-clockwise. Where does $\boldsymbol{x} = \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$ go?

Solution. $\boldsymbol{R}\boldsymbol{x} = \left(\begin{smallmatrix} 0\cdot1-1\cdot0 \\ 1\cdot1+0\cdot0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)$: the arrow pointing right becomes the arrow pointing up, exactly a quarter turn. The columns of $\boldsymbol{R}$ are where the two axis vectors land.

3.3.2 Matrix–matrix multiplication

Multiplying two matrices means *composing* their transformations: $\boldsymbol{A}\boldsymbol{B}$ is "do $\boldsymbol{B}$, then $\boldsymbol{A}$". Entry (i, j) of the product is the dot product of row i of $\boldsymbol{A}$ with column j of $\boldsymbol{B}$.

Multiplying two matrices

Let $\boldsymbol{A} = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$ and $\boldsymbol{B} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ (which swaps components).

Solution.

$$\boldsymbol{A}\boldsymbol{B} = \begin{pmatrix} 1\cdot0+2\cdot1 & 1\cdot1+2\cdot0 \\ 3\cdot0+4\cdot1 & 3\cdot1+4\cdot0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}.$$

Right-multiplying by $\boldsymbol{B}$ swapped the *columns* of $\boldsymbol{A}$.

Common pitfall

Matrix multiplication is **not commutative**: in general $\boldsymbol{AB} \neq \boldsymbol{BA}$. Order encodes the order of operations ("rotate then stretch" differs from "stretch then rotate"). Also check dimensions: an $m \times n$ times an $n \times p$ gives $m \times p$; the inner dimensions must match.

The **identity matrix** $\boldsymbol{I}$ (ones on the diagonal, zeros elsewhere) leaves every vector unchanged: $\boldsymbol{Ix} = \boldsymbol{x}$. The **transpose** $\boldsymbol{A}^\top$ flips rows and columns; note that $\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{u}^\top \boldsymbol{v}$, tying the dot product to matrix notation.

Algebra of matrices

Matrix multiplication is *associative* $(\boldsymbol{AB})\boldsymbol{C} = \boldsymbol{A}(\boldsymbol{BC})$ and *distributive* $\boldsymbol{A}(\boldsymbol{B}+\boldsymbol{C}) = \boldsymbol{AB} + \boldsymbol{AC}$, but *not commutative*. Transpose reverses a product: $(\boldsymbol{AB})^\top = \boldsymbol{B}^\top \boldsymbol{A}^\top$. And $\boldsymbol{AI} = \boldsymbol{IA} = \boldsymbol{A}$. These rules let us manipulate matrix expressions just like numbers — *except* that order must be preserved.

Matrices — do together

Let $\boldsymbol{A} = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$ (a horizontal shear) and $\boldsymbol{x} = \left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$.

1. Compute $\boldsymbol{Ax}$.

   Answer. $\left(\begin{smallmatrix} 1+2 \\ 0+1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 3 \\ 1 \end{smallmatrix}\right)$.
2. Compute $\boldsymbol{AA}$ (apply the shear twice).

   Answer. $\left(\begin{smallmatrix} 1 & 4 \\ 0 & 1 \end{smallmatrix}\right)$.
3. Does $\boldsymbol{A}$ change area? (Find $\det \boldsymbol{A}$.)

   Answer. $\det = 1 \cdot 1 - 2 \cdot 0 = 1$: a shear preserves area.

3.4 Linear systems and the inverse (≈*25 min*)

A system of linear equations is compactly a single matrix equation.

Definition

The system $\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases}$ is written $\boldsymbol{Ax} = \boldsymbol{b}$. Solving it means finding the vector $\boldsymbol{x}$ that $\boldsymbol{A}$ maps to $\boldsymbol{b}$.

Intuition

Two readings again. *Row picture:* each equation is a line, and a solution is where the lines cross. *Column picture:* we ask which combination of $\boldsymbol{A}$'s columns produces $\boldsymbol{b}$. Both views matter; the column picture is the one that generalises to high dimensions.

Solving a 2×2 system

Solve $x + y = 3, \quad x - y = 1$.

Solution. Add the equations: $2x = 4 \Rightarrow x = 2$. Then $y = 3 - 2 = 1$. The two lines cross at $(2, 1)$.

When a transformation can be undone, it has an inverse.

Definition

The **inverse** $\boldsymbol{A}^{-1}$ satisfies $\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}$; it exists only for square matrices with nonzero *determinant*. For a 2×2 matrix,

$$\boldsymbol{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \det \boldsymbol{A} = ad - bc, \quad \boldsymbol{A}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

When $\boldsymbol{A}^{-1}$ exists, the system has the unique solution $\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b}$.

Intuition

The determinant measures how much the transformation *scales area* (or volume). If $\det \boldsymbol{A} = 0$, the transformation squashes space flat — it collapses a dimension — so information is lost and it cannot be undone. A zero determinant is the algebraic signature of "this system may have no solution, or infinitely many".

A 2×2 inverse

Invert $\boldsymbol{A} = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$.

Solution. $\det \boldsymbol{A} = 1 \cdot 4 - 2 \cdot 3 = -2 \neq 0$, so

$$\boldsymbol{A}^{-1} = \frac{1}{-2}\begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix}.$$

One can check $\boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{I}$.

Determinant and inverse facts

- $\det(\boldsymbol{A}\boldsymbol{B}) = \det \boldsymbol{A} \det \boldsymbol{B}$, and $\det(\boldsymbol{A}^{\mathsf{T}}) = \det \boldsymbol{A}$.
- $\boldsymbol{A}$ is invertible $\iff$ $\det \boldsymbol{A} \neq 0$ $\iff$ its columns are linearly independent $\iff$ $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ has a unique solution for every $\boldsymbol{b}$. (These are all the same statement.)
- Product rule for inverses: $(\boldsymbol{A}\boldsymbol{B})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$ (undo the last operation first).

For larger systems we solve by **Gaussian elimination**: use one equation to cancel a variable from the others, repeat, then back-substitute. It is the same idea as the 2×2 case, scaled up.

Elimination on a 3×3 system

Solve $x + y + z = 6, \quad 2x - y + z = 3, \quad x + 2y - z = 2.$

Solution. Subtract twice the first equation from the second, and the first from the third, to remove x:

$$(2x-y+z)-2(x+y+z) = 3-12 \Rightarrow 3y+z = 9, \qquad (x+2y-z)-(x+y+z) = 2-6 \Rightarrow y-2z = -4.$$

From the first, $z = 9 - 3y$; substitute into the second: $y - 2(9 - 3y) = -4 \Rightarrow 7y = 14 \Rightarrow y = 2$. Then $z = 3$ and $x = 6 - 2 - 3 = 1$. Solution: $(x, y, z) = (1, 2, 3)$.

3.5 Independence, rank, and dimension (≈*25 min*)

These ideas answer: *how much genuinely different information is in a set of vectors?*

Definition

Vectors are **linearly independent** if none is a linear combination of the others (no redundancy). The **span** of a set of vectors is all linear combinations you can build from them — the space they "reach". A **basis** is a minimal set of independent vectors whose span is the whole space; the number of them is the **dimension**.

Definition

The **rank** of a matrix is the number of linearly independent columns (equivalently, rows). It is the dimension of the space the matrix can reach — the true number of "effective directions" in the data.

Intuition

Independence is redundancy-detection. The vectors $\binom{1}{2}$ and $\binom{2}{4}$ point the same way — the second adds nothing — so together they span only a line (rank 1). By contrast $\binom{1}{0}$ and $\binom{0}{1}$ reach the whole plane (rank 2). Rank is the honest count of dimensions your data actually occupies, which may be far fewer than the number of columns.

Quick tests for independence

- Two vectors are dependent exactly when one is a scalar multiple of the other (they point along the same line).
- For a square matrix, columns are independent $\iff$ $\det \neq 0$.
- **Dimension bound:** any set of more than n vectors in $\mathbb{R}^n$ is automatically dependent — you cannot have 3 independent directions in a plane.

Finding the rank

What is the rank of the columns of $\boldsymbol{M} = \left(\begin{smallmatrix} 1 & 3 & 2 \\ 2 & 6 & 5 \end{smallmatrix}\right)$?

Solution. The second column $\binom{3}{6}$ is exactly $3\times$ the first, so it adds no new direction. The third, $\binom{2}{5}$, is not a multiple of the first (the ratios $2/1$ and $5/2$ differ), so it does. Two independent directions in total: rank $\boldsymbol{M} = 2$. (It cannot exceed 2 anyway, since the columns live in $\mathbb{R}^2$.)

Rank & independence — do together

1. Are $\binom{1}{2}$ and $\binom{2}{4}$ independent?

   Answer. No — the second is twice the first (rank 1).
2. Are $\binom{1}{0}$ and $\binom{1}{1}$ independent?

   Answer. Yes — not multiples; $\det\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right) = 1 \neq 0$.
3. Can three vectors in $\mathbb{R}^2$ ever be independent?

   Answer. No — at most 2 independent directions exist in a plane.

Why it matters for cognitive science

This is exactly what we mean by the *dimensionality* of neural or behavioural data. A recording of 100 neurons lives in $\mathbb{R}^{100}$, but their activity during a task often varies along only a handful of independent directions — the data have low rank. Finding those few directions is *dimensionality reduction*, and the principled way to do it is Principal Component Analysis, which is exactly where the next session begins. The covariance matrix from Session 2 will supply the directions.

Session recap

Vectors are points/arrows/feature-lists, combined by *linear combination*. The *dot product* encodes length, distance, angle, and similarity, with *orthogonality* (dot product zero) as the geometric form of "unrelated". A *matrix* is both a data table and a transformation; *matrix multiplication* composes transformations and is not commutative. Linear systems $\boldsymbol{Ax} = \boldsymbol{b}$ are solved via the *inverse* when $\det \boldsymbol{A} \neq 0$, and *rank/independence/basis* measure the true dimensionality of data. Next session: the special directions of a transformation — *eigenvectors* — and their flagship application, *PCA*.

Chapter 4

Linear Algebra II — Eigenvectors, PCA, and Dynamics

Where this session is going

This is the payoff session, where the two halves of the course meet. We introduce the *special directions* of a transformation — its *eigenvectors* and *eigenvalues* — lead-ing with geometry, not algebra. We then meet the *covariance matrix* again and use its eigenvectors to build **Principal Component Analysis**, the standard tool for reducing and visualising high-dimensional neural and behavioural data. We sketch the *SVD* as the general engine behind PCA, and close by showing how eigenvalues govern the stability of *linear dynamical systems* — the link to models of neural population dynamics.

4.1 Eigenvectors and eigenvalues (≈*35 min*)

Most vectors get knocked off their direction when a matrix acts on them. A few special ones do not.

Definition

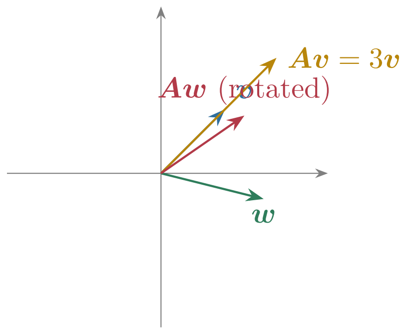
A nonzero vector $\boldsymbol{v}$ is an **eigenvector** of a square matrix $\boldsymbol{A}$ if $\boldsymbol{A}$ only *stretches* it, without changing its direction:

$$\boldsymbol{A}\boldsymbol{v} = \lambda\, \boldsymbol{v},$$

where the number λ is the **eigenvalue** — the stretch factor along that direction.

Intuition

Think of the transformation as flowing space around. Almost every arrow gets rotated as well as stretched. But along an eigenvector, the transformation is pure stretch: the arrow stays on its own line, only getting longer, shorter, or flipped (if $\lambda < 0$). Eigenvectors are the "grain of the wood" — the natural axes along which the transformation is simplest. Along them, a complicated matrix behaves like plain multiplication by a number.



To find eigenvalues we solve the **characteristic equation** $\det(\boldsymbol{A} - \lambda\boldsymbol{I}) = 0$; each root λ then gives its eigenvectors by solving $(\boldsymbol{A} - \lambda\boldsymbol{I})\boldsymbol{v} = \mathbf{0}$.

Eigenvalues and eigenvectors of a 2×2

Find the eigenpairs of $\boldsymbol{A} = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$.

Solution. The characteristic equation:

$$\det\begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1 = 0 \;\Rightarrow\; (2-\lambda)^2 = 1 \;\Rightarrow\; \lambda = 3 \text{ or } \lambda = 1.$$

For $\lambda = 3$: $(\boldsymbol{A} - 3\boldsymbol{I})\boldsymbol{v} = \left(\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right)\boldsymbol{v} = \mathbf{0} \Rightarrow -v_1 + v_2 = 0$, so $\boldsymbol{v}_1 = \left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$. For $\lambda = 1$: $v_1 + v_2 = 0$, so $\boldsymbol{v}_2 = \left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$. Notice the two eigenvectors are *orthogonal*: $\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = 1 - 1 = 0$.

Two free checks on your eigenvalues

For any square matrix, the eigenvalues satisfy

$$\sum_i \lambda_i = \operatorname{tr}(\boldsymbol{A}) \text{ (sum of the diagonal)}, \qquad \prod_i \lambda_i = \det(\boldsymbol{A}).$$

For the example above: $\lambda_1 + \lambda_2 = 3 + 1 = 4 = \operatorname{tr}(\boldsymbol{A})$ and $\lambda_1\lambda_2 = 3 \cdot 1 = 3 = \det(\boldsymbol{A})$. Always use these to catch arithmetic slips before going further.

Diagonalization — and why eigenvalues make powers easy

If $\boldsymbol{A}$ has a full set of independent eigenvectors, collect them as columns of $\boldsymbol{P}$ and the eigenvalues on the diagonal of $\boldsymbol{D}$; then $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$. In the eigenvector coordinates the matrix is just the diagonal of stretches. This makes repeated application trivial:

$$\boldsymbol{A}^k = \boldsymbol{P}\boldsymbol{D}^k\boldsymbol{P}^{-1}, \qquad \boldsymbol{D}^k = \operatorname{diag}(\lambda_1^k, \ldots, \lambda_n^k).$$

So the long-run behaviour of $\boldsymbol{A}^k$ is governed entirely by the eigenvalues — exactly what we need for dynamical systems below.

Eigenvalues — do together

Let $\boldsymbol{A} = \left(\begin{smallmatrix} 3 & 0 \\ 0 & -2 \end{smallmatrix}\right)$ (already diagonal).

1. What are its eigenvalues and eigenvectors?

   Answer. $\lambda = 3$ with $\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$, and $\lambda = -2$ with $\left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)$.
2. Check the trace and determinant.

   Answer. $\operatorname{tr} = 1$ and $3 + (-2) = 1$; $\det = -6$ and $3 \cdot (-2) = -6$. ✓
3. What does $\boldsymbol{A}$ do to the vector $\left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$?

Answer. $\binom{3}{-2}$: stretch x by 3, flip and stretch y by 2.

Key idea

Symmetric matrices are special. If $\boldsymbol{A} = \boldsymbol{A}^\mathsf{T}$ (symmetric — as every covariance matrix is), then its eigenvalues are real and its eigenvectors are mutually orthogonal. The transformation is then just a set of independent stretches along perpendicular axes. This clean structure is precisely what makes PCA work.

A symmetric matrix is **positive-definite** when all its eigenvalues are positive; geometrically it stretches (never flips) along every axis. Covariance matrices are positive-(semi)definite because variances cannot be negative — a fact worth stating, as it guarantees PCA's directions have non-negative "amounts of variance".

4.2 Principal Component Analysis (≈*45 min*)

Now the flagship application, unifying statistics and linear algebra.

4.2.1 The problem

High-dimensional data are hard to see and often redundant: many measured variables are correlated, so the cloud of data points really lies near a lower-dimensional subspace. PCA finds the few directions that capture the most variation, so we can describe, visualise, and compress the data with little loss.

Key idea

PCA in one sentence: the principal components are the *eigenvectors of the covariance matrix*, and each eigenvalue is the *amount of variance* the data have along that direction. The first principal component is the single direction of greatest spread; the next is the direction of greatest remaining spread orthogonal to the first; and so on.

4.2.2 The recipe

1. **Centre** the data: subtract each variable's mean, so the cloud sits at the origin.
2. Form the **covariance matrix** $\boldsymbol{C}$ of the centred data (the same object from Session 2, now for many variables at once).
3. Compute the **eigenvectors and eigenvalues** of $\boldsymbol{C}$.
4. Order eigenvectors by eigenvalue, largest first: these are principal components $1, 2, 3, \ldots$
5. **Project** the data onto the top few components to reduce dimension.

Intuition

Picture an elongated, tilted cloud of points. PCA finds the long axis of the cloud (most variance — the first component) and the short axis perpendicular to it (the second). Rotating onto these axes "straightens" the cloud so its directions become uncorrelated. Keeping only the long axis is dimensionality reduction: we describe each point by its position along the direction that matters most.

PCA on a 2×2 covariance matrix

Suppose two mean-centred variables have covariance matrix $\boldsymbol{C} = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$.

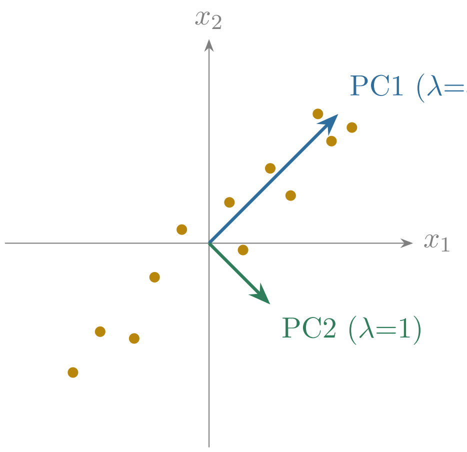
Solution. We already found its eigenpairs above: $\lambda_1 = 3$ with direction $\left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$, and $\lambda_2 = 1$ with direction $\left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$. So the **first principal component** points along $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$ (the two variables rise together) and carries variance 3; the second, orthogonal, carries variance 1. The fraction of total variance explained by the first component is

$$\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{3}{3+1} = 75\%.$$

Reducing these two variables to their first component keeps 75% of the variance in a single number.

Reading the eigenvalues of a covariance matrix

- The **total variance** equals the sum of the eigenvalues, which is also the trace of $\boldsymbol{C}$ (the sum of the individual variances): $\sum_i \lambda_i = \operatorname{tr}(\boldsymbol{C})$.
- The **proportion of variance explained** by component j is $\lambda_j / \sum_i \lambda_i$. Plotting these in decreasing order gives the **scree plot**; a sharp "elbow" suggests how many components to keep.
- Because $\boldsymbol{C}$ is symmetric and positive-semidefinite, all $\lambda_i \geq 0$ — variances are never negative.



Why it matters for cognitive science

PCA is everywhere in cognitive science and neuroscience. It reduces recordings of hundreds of neurons to a few "latent" dimensions along which population activity evolves during a task (neural state-space analysis). It compresses fMRI voxels and EEG channels, extracts principal components of face or object images ("eigenfaces"), and summarises questionnaire batteries into a few underlying factors (closely related to factor analysis). Whenever someone says a system is "effectively low-dimensional", PCA is usually how they found out.

Common pitfall

PCA is a *linear* method and is sensitive to *scaling*: a variable measured in large units (e.g. reaction time in ms) can dominate the components purely because of its units. Standardising variables (or using the correlation matrix instead of the covariance matrix) is often necessary. PCA also finds directions of variance, which are not always the directions that matter for a task — variance is not the same as relevance.

PCA — do together

After PCA on some neural data, the (already diagonalised) covariance has eigenvalues $\lambda = (6, 3, 1)$ along three orthogonal components.

1. What is the total variance?

   Answer. $6 + 3 + 1 = 10$.
2. What fraction does the first component explain?

   Answer. $6/10 = 60\%$.
3. How many components to keep for at least 85% of the variance?

   Answer. Two: $(6 + 3)/10 = 90\% \geq 85\%$.

4.2.3 SVD — the general engine

The **Singular Value Decomposition** writes *any* matrix as $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$: a rotation, a set of non-negative stretches (the *singular values* on $\boldsymbol{\Sigma}$), and another rotation. PCA is SVD applied to the centred data matrix; the singular values squared are the variances (eigenvalues of the covariance matrix). Students need only the idea: *every linear transformation is a rotation, then a scaling along perpendicular axes, then another rotation*, and SVD is how we find those axes for non-square data.

4.3 Linear dynamical systems: eigenvalues over time (≈*20 min*)

Eigenvalues also predict the *future*. Many models describe how a state $\boldsymbol{x}$ evolves under repeated application of a matrix, or continuously in time.

Key idea

Discrete time: if $\boldsymbol{x}_{t+1} = \boldsymbol{A}\boldsymbol{x}_t$, then along each eigenvector the state is multiplied by λ each step. The system *shrinks to zero* (is stable) if every $|\lambda| < 1$, and *blows up* if some $|\lambda| > 1$.

Continuous time: if $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}$, the solution grows like $e^{\lambda t}$ along each eigenvector. The system is *stable* (returns to equilibrium) if every eigenvalue has *negative real part*, and unstable otherwise; imaginary parts produce oscillations.

Intuition

Eigenvalues are the "growth rates" of the natural modes of a system. Decompose any starting state into eigenvector components; each then evolves independently, growing or decaying at its own rate. The largest eigenvalue eventually dominates, setting the long-run behaviour. This is why the eigenvalues alone tell you whether a dynamical system settles, oscillates, or explodes.

Reading stability off the eigenvalues

A discrete system $\boldsymbol{x}_{t+1} = \boldsymbol{A}\boldsymbol{x}_t$ has eigenvalues $\lambda_1 = 0.5$ and $\lambda_2 = 1.2$. What happens over time?

Solution. Along the first eigenvector the state is multiplied by 0.5 each step, so that component decays to zero. Along the second it is multiplied by 1.2, growing by 20% each step. Since one $|\lambda| > 1$, the system is **unstable**: almost every starting state eventually grows without bound, aligning with the λ_2 direction. It would be stable only if *both* $|\lambda| < 1$.

Dynamics — do together

1. A continuous system $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}$ has eigenvalues -1 and -3. Stable or unstable?

   Answer. Stable — both real parts negative, so every mode decays.
2. Another has eigenvalues -2 and $+0.5$.

   Answer. Unstable — the $+0.5$ mode grows like $e^{0.5t}$.
3. A discrete system has eigenvalues 0.9 and 0.4.

   Answer. Stable — both $|\lambda| < 1$, so the state shrinks to zero.

Why it matters for cognitive science

Recurrent neural networks and models of neural population dynamics are exactly systems of the form $\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x}$ (plus inputs and nonlinearities). Whether a memory persists, an activity pattern decays, or a network oscillates is governed by the eigenvalues of its connectivity matrix. This closes the loop with the differential-equations material students meet elsewhere in the programme: the qualitative behaviour of a linear system is read off from its spectrum.

Session recap

Eigenvectors are the directions a matrix only stretches, with the stretch factor given by the *eigenvalue*; *symmetric* matrices (like every covariance matrix) have real eigenvalues and orthogonal eigenvectors. *PCA* exploits this: the eigenvectors of the covariance matrix are the principal components, and the eigenvalues are the variances along them — giving a principled way to reduce and visualise high-dimensional neural and behavioural data. *SVD* is the general decomposition behind it. Finally, in *dynamical systems* the eigenvalues set the growth or decay of each mode and thus the stability of models of neural dynamics. Next session returns to probability, now armed with vectors and matrices, to build the modeller's statistical toolkit.

Chapter 5

Advanced Probability & Statistics — The Modeller's Toolkit

Where this session is going

This session is a recap-and-elevation for students who already have some mathematics: it fuses the probability of Sessions 1–2 with the linear algebra of Sessions 3–4 into the tools actually used to build models. We deepen *Bayesian inference*; extend to *several variables at once* and the *multivariate Gaussian*; state the one estimation principle behind almost all model fitting, *maximum likelihood*; unify regression and its cousins as the *Generalized Linear Model*; confront *overfitting* through *model comparison*; introduce *information theory*; and close with a glimpse of *Markov chains and sampling*.

5.1 Bayesian inference, in depth (≈*35 min*)

In Session 1 Bayes' theorem updated the probability of a single hypothesis. As a *framework*, it updates a whole distribution over a continuous parameter.

Key idea

For a parameter θ and data D,

$$\underbrace{p(\theta \mid D)}_{\text{posterior}} = \frac{\overbrace{p(D \mid \theta)}^{\text{likelihood}}\ \overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(D)}_{\text{evidence}}} \propto p(D \mid \theta)\, p(\theta).$$

We start with a *prior* distribution over θ, weight it by how well each θ predicts the data (the *likelihood*), and renormalise to get the *posterior* — our full updated belief.

Intuition

The Bayesian answer is never a single number but a *distribution*: it carries its own uncertainty. A sharp posterior means the data pinned θ down; a broad one means we are still unsure. Two summaries are common: the **posterior mean** (the average belief) and the **MAP estimate** (the single most probable θ, the peak of the posterior).

Certain prior–likelihood pairs are **conjugate**: the posterior stays in the same family, so updating is just arithmetic on the parameters. The Beta–Binomial pair is the classic teaching example.

Estimating a probability the Bayesian way

We want the probability θ that a participant detects a faint stimulus. We put a uniform prior $\theta \sim \mathrm{Beta}(1,1)$ (all values equally likely) and observe 8 detections in 10 trials.

Solution. For a $\mathrm{Beta}(a,b)$ prior and h successes in $h+t$ trials, the posterior is $\mathrm{Beta}(a+h,\ b+t)$. Here:

$$\theta \mid D \sim \mathrm{Beta}(1+8,\ 1+2) = \mathrm{Beta}(9,3).$$

Its **posterior mean** is $\dfrac{9}{9+3} = 0.75$, while its **MAP** (mode) is $\dfrac{9-1}{9+3-2} = \dfrac{8}{10} = 0.80$, which coincides with the raw frequency. The prior gently pulls the estimate away from the extreme, and the posterior's width honestly reports that 10 trials leave real uncertainty.

Two facts that make Bayes practical

- **Sequential updating:** today's posterior is tomorrow's prior. Processing data all at once or one point at a time gives the *same* posterior — so beliefs can be updated online, as evidence arrives.
- **Credible interval:** a Bayesian "95% credible interval" is a range holding 95% of the posterior probability. Unlike a confidence interval, it *does* permit the natural statement "there is a 95% probability the parameter lies in here", because θ is treated as uncertain.

Bayesian updating — do together

Start from a uniform $\mathrm{Beta}(1,1)$ prior on a detection probability θ.

1. After 3 hits in 3 trials, what is the posterior?

   Answer. $\mathrm{Beta}(1+3,\ 1+0) = \mathrm{Beta}(4,1)$.
2. What is its posterior mean?

   Answer. $4/(4+1) = 0.8$.
3. You then see 1 hit in the next 2 trials. Update again.

   Answer. $\mathrm{Beta}(4+1,\ 1+1) = \mathrm{Beta}(5,2)$, mean $5/7 \approx 0.71$. (Same as updating from all $4/5$ hits at once.)

Why it matters for cognitive science

Bayesian inference is the dominant formal language of contemporary cognitive modeling: models of perception as inference, of cue combination, of learning and generalisation, and of "the Bayesian brain" all cast cognition as computing posteriors. The prior encodes expectations; the likelihood encodes sensory evidence; behaviour reflects the posterior. Understanding this template lets students read a large fraction of the modeling literature.

Common pitfall

The prior is a modeling choice, not a nuisance to hide. With little data it matters; with abundant data the likelihood dominates and the posterior becomes insensitive to reasonable priors. Students should report priors explicitly and check that conclusions are robust to them.

5.2 Many variables at once: the multivariate Gaussian (≈*25 min*)

Real data have many variables that must be modelled *jointly*. This is where probability and linear algebra become the same subject.

Definition

For several variables, the **joint** distribution $p(x, y)$ describes them together; the **marginal** $p(x) = \sum_y p(x, y)$ describes one alone (summing/integrating out the rest); the **conditional** $p(x \mid y)$ describes one given the value of another. They are linked by $p(x, y) = p(x \mid y)\, p(y)$.

Key idea

The **multivariate Gaussian** $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ generalises the bell curve to $\mathbb{R}^n$. It is defined by a *mean vector* $\boldsymbol{\mu}$ (the centre) and a *covariance matrix* $\boldsymbol{\Sigma}$ (the shape): the diagonal holds the variances, the off-diagonal the covariances. Its contours of equal probability are ellipses whose axes are the *eigenvectors* of $\boldsymbol{\Sigma}$ and whose widths are set by the *eigenvalues* — exactly the PCA picture from Session 4.

Intuition

The covariance matrix *is* the object that needs both toolkits: statistics supplies its entries (variances and covariances), and linear algebra reads its shape (eigenvectors and eigenvalues give the orientation and spread of the data ellipse). This is the single most important unification in the course — the multivariate Gaussian is where "covariance", "eigenvector", and "principal component" turn out to be three views of one thing.

Why the Gaussian is so convenient

- **Closed under marginals and conditionals:** if a set of variables is jointly Gaussian, then each subset is Gaussian, and conditioning on some of them leaves the rest Gaussian too. (Almost no other distribution is so well-behaved.)
- **For Gaussians, uncorrelated $\Rightarrow$ independent.** In general zero covariance need not mean independence — but for jointly Gaussian variables it does. So a diagonal covariance matrix means fully independent components.
- **Linear maps stay Gaussian:** if $\boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ then $\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b} \sim \mathcal{N}(\boldsymbol{A}\boldsymbol{\mu} + \boldsymbol{b},\ \boldsymbol{A}\boldsymbol{\Sigma}\boldsymbol{A}^\top)$. This is exactly what PCA uses when it rotates the data onto its components.

Why it matters for cognitive science

Multivariate Gaussians underlie models of neural noise correlations, Gaussian generative models and linear discriminant analysis for classification, Kalman filters for tracking and motor control, and the latent-variable models (e.g. factor analysis, probabilistic PCA) used to describe population activity. Whenever "noise is correlated across neurons", a covariance matrix is doing the work.

5.3 Maximum likelihood: the fitting principle (≈*25 min*)

How do we choose parameters to fit a model? One idea unifies nearly all of it.

Definition

The **maximum likelihood estimate** (MLE) is the parameter value that makes the observed data most probable:

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta}\ p(D \mid \theta) = \arg\max_{\theta}\ \sum_i \log p(x_i \mid \theta),$$

where we maximise the *log*-likelihood (a sum, easier than a product).

Intuition

Likelihood reverses the usual question. A distribution asks "given the parameters, how probable is each possible dataset?" Likelihood fixes the data we actually saw and asks "which parameters would have made *this* most plausible?" MLE simply picks that best-explaining parameter. Taking the log turns the product over independent data points into a sum, which is what we actually optimise.

Key idea

MLE unifies methods students already know. For Gaussian data, the MLE of the mean is just the *sample mean*. Fitting a regression line by *least squares* is exactly maximum likelihood under Gaussian noise. So "minimise squared error" and "maximise likelihood" are the same procedure in disguise, and the same idea scales up to fitting essentially any probabilistic model, including in machine learning where the log-likelihood becomes the "loss" to minimise.

Deriving the obvious estimate the formal way

A participant succeeds on k of n independent trials. Find the MLE of the success probability p.

Solution. The log-likelihood is $\ell(p) = k \ln p + (n-k)\ln(1-p)$. Differentiate and set to zero:

$$\ell'(p) = \frac{k}{p} - \frac{n-k}{1-p} = 0 \implies k(1-p) = (n-k)p \implies \hat{p} = \frac{k}{n}.$$

Maximum likelihood recovers the natural estimate — the observed success rate — and the same machinery gives estimates when the answer is *not* obvious.

Likelihood — do together

1. A neuron responds on 12 of 40 trials. What is the MLE of its response probability?

   Answer. $12/40 = 0.3$.
2. Two coin models: A says $p = 0.5$, B says $p = 0.7$. You see 7 heads in 10 flips. Which assigns higher likelihood?

   Answer. B: $0.7^7 0.3^3 \approx 0.00222$ vs. A: $0.5^{10} \approx 0.00098$.
3. Why maximise the *log*-likelihood rather than the likelihood itself?

   Answer. The log turns products into sums and does not move the maximum (it is an increasing function).

Why it matters for cognitive science

Almost every model students will fit — psychometric functions, reinforcement- learning models of choice, drift-diffusion models of reaction time, neural encoding models — is fit by maximising a likelihood (or its Bayesian cousin, the posterior). MLE is the workhorse that turns a model into numbers you can compare to data.

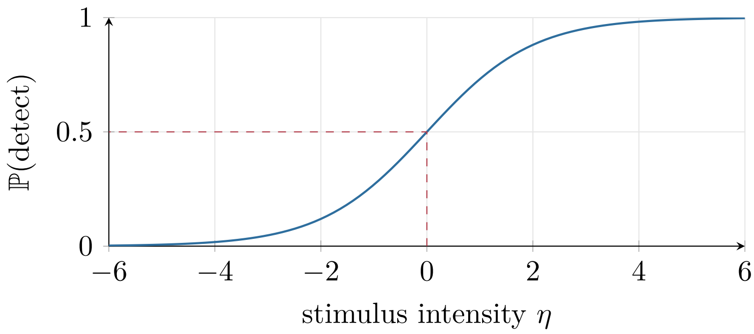
5.4 The Generalized Linear Model (≈*25 min*)

Regression assumed a straight line and Gaussian noise. Many outcomes are binary (correct/incorrect) or counts (spikes). The GLM extends regression to these while keeping its linear core.

Definition

A **Generalized Linear Model** keeps a linear predictor $\eta = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$ but passes it through a **link function** so the output matches the data type. For binary outcomes, **logistic regression** uses the logistic (sigmoid) link:

$$\mathbb{P}(y = 1 \mid x) = \sigma(\eta) = \frac{1}{1 + e^{-\eta}}.$$



Intuition

The link function is a translator. The linear part η can be any number from $-\infty$ to ∞, but a probability must live in $[0, 1]$ and a count must be non-negative. The sigmoid squashes the whole real line into $[0, 1]$, giving the familiar S-shaped *psychometric function*: as stimulus intensity rises, the probability of a "yes" response climbs smoothly from 0 to 1.

Why it matters for cognitive science

The GLM is the quiet backbone of quantitative psychology and neuroscience. Logistic regression *is* the psychometric function relating stimulus to detection probability. Poisson regression models spike counts as a function of stimuli or covariates (the "linear–nonlinear–Poisson" encoding model). The same framework yields decoding models that read stimuli off neural activity. Master the GLM and a huge swath of analysis methods become one familiar object.

What logistic-regression coefficients mean

Inverting the sigmoid shows the linear predictor *is* the log-odds:

$$\ln \frac{\mathbb{P}(y = 1)}{\mathbb{P}(y = 0)} = \beta_0 + \beta_1 x_1 + \ldots$$

So each coefficient β_j is the change in *log-odds* per unit of x_j, and e^{β_j} is the **odds ratio** — the multiplicative effect on the odds. Positive β pushes the probability up, negative pushes it down, and $\beta = 0$ means no effect (odds ratio $= 1$).

GLM — do together

A logistic model of "detected" has $\eta = \beta_0 + \beta_1 x$ with $\beta_0 = 0$.

1. At $\eta = 0$, what is the detection probability?

   Answer. $\sigma(0) = 1/2$.

2. As x (hence η) grows large, what does the probability approach?

   Answer. $\to 1$ (the sigmoid saturates).

3. If $\beta_1 = \ln 2$, by what factor do the odds of detection change per unit of x?

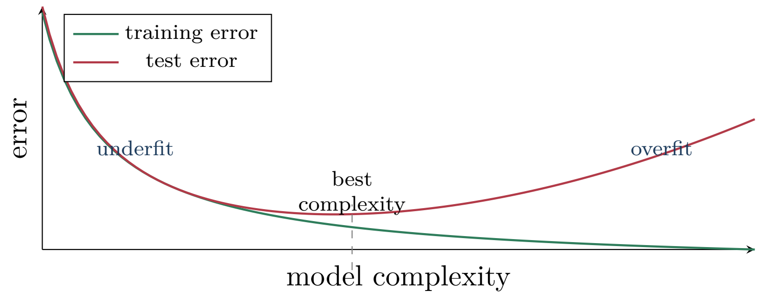
   Answer. $e^{\beta_1} = 2$: the odds double.

5.5 Overfitting and model comparison (≈*25 min*)

A model that fits the training data perfectly may predict new data terribly. Guarding against this is arguably the most transferable skill in the course.

Key idea

The bias–variance tradeoff. Too simple a model *underfits* (high bias: it misses real structure). Too complex a model *overfits* (high variance: it chases noise and fails to generalise). Good modeling finds the middle: enough flexibility to capture the signal, not so much that it memorises the noise.



Key idea

Two practical tools.

Cross-validation: hold out part of the data, fit on the rest, and measure prediction error on the held-out part. This directly estimates how the model generalises — the gold standard.

Information criteria: score a model by its fit *penalised* for complexity, $\mathrm{AIC} = 2k - 2\ln\hat{L}$ and $\mathrm{BIC} = k\ln n - 2\ln\hat{L}$, where k is the number of parameters and $\hat{L}$ the maximised likelihood. Lower is better; both reward fit and punish extra parameters.

AIC and BIC can disagree

Two models are fit to $n = 100$ data points. Model 1 has $k = 2$ parameters and maximised log-likelihood $\ln\hat{L} = -50$; Model 2 has $k = 4$ and $\ln\hat{L} = -47$ (it fits a little better, as a more flexible model must). Which is preferred?

Solution.

$$\mathrm{AIC}_1 = 2(2) - 2(-50) = 104, \qquad \mathrm{AIC}_2 = 2(4) - 2(-47) = 102 \;\Rightarrow\; \text{AIC prefers Model 2.}$$

$$\mathrm{BIC}_1 = 2\ln 100 - 2(-50) \approx 109.2, \qquad \mathrm{BIC}_2 = 4\ln 100 - 2(-47) \approx 112.4 \;\Rightarrow\; \text{BIC prefers Model 1.}$$

BIC's heavier penalty ($\ln n \approx 4.6$ per parameter vs. AIC's 2) makes it favour the simpler model. When criteria disagree, cross-validation is the tie-breaker.

Model comparison — do together

1. Adding parameters, training error always goes down. Does that mean the bigger model is better?

   Answer. No — it may be overfitting; judge on held-out data.

2. Between two models with equal test error, which do you keep?
   Answer. The simpler one (parsimony / lower variance).
3. Which penalises complexity more for $n > 7$, AIC or BIC?
   Answer. BIC, since $\ln n > 2$ once $n \geq 8$.

Common pitfall

Never judge a model only by how well it fits the data it was trained on — more parameters *always* fit training data better. The honest question is out-of-sample prediction. Comparing models on training error alone is the single most common mistake in applied modeling.

5.6 A first taste of information theory (≈*20 min*)

Information theory gives a precise language for uncertainty and surprise, widely used in neural coding and machine learning.

Definition

The **entropy** of a distribution measures its average uncertainty (in bits when using $\log_2$):

$$H(X) = -\sum_x p(x) \log_2 p(x).$$

The **Kullback–Leibler divergence** measures how different a distribution q is from a reference p (always ≥ 0, and 0 only when they are equal): $D_{\mathrm{KL}}(p \,\|\, q) = \sum_x p(x) \log_2 \frac{p(x)}{q(x)}$. The **mutual information** $I(X;Y)$ measures how much knowing Y reduces uncertainty about X.

Intuition

Entropy is "how surprised do I expect to be?" A fair coin has entropy 1 bit (maximally uncertain); a two-headed coin has entropy 0 (no surprise). KL divergence is the extra cost of using the wrong distribution to describe reality. Mutual information is the shared information between two variables — the information-theoretic cousin of correlation, but sensitive to *any* dependence, not just linear.

Entropy of a biased coin

A coin lands heads with probability 0.25. What is its entropy?

Solution.

$$H = -0.25 \log_2 0.25 - 0.75 \log_2 0.75 = 0.25(2) + 0.75(0.415) \approx 0.5 + 0.311 = 0.81 \text{ bits}.$$

Less than the 1 bit of a fair coin: the bias makes outcomes more predictable.

Three facts about entropy and its relatives

- **Maximum entropy:** among distributions on m outcomes, the uniform one has the largest entropy, $\log_2 m$ bits (maximal uncertainty).
- **Cross-entropy = entropy + KL:** $H(p,q) = H(p) + D_{\mathrm{KL}}(p\|q)$. Since KL ≥ 0, using the wrong model q always costs extra bits — which is exactly why minimising cross-entropy loss trains a classifier toward the true distribution.

- **Mutual information** is symmetric and non-negative: $I(X;Y) = H(X) - H(X \mid Y) \geq 0$, and equals 0 exactly when X and Y are independent.

Information — do together

1. What is the entropy of a fair six-sided die?

   Answer. $\log_2 6 \approx 2.585$ bits.
2. Which has more entropy: a fair coin or a coin with $p = 0.9$?

   Answer. The fair coin (1 bit vs. ≈ 0.47 bits).
3. If knowing Y makes X certain, what is $I(X;Y)$ in terms of $H(X)$?

   Answer. $I(X;Y) = H(X)$ — Y removes all of X's uncertainty.

Why it matters for cognitive science

Information theory quantifies how much a neuron or population tells you about a stimulus (mutual information between stimulus and response), underlies the "efficient coding" hypothesis that sensory systems maximise information, and supplies the loss functions of machine learning — cross-entropy loss is exactly a KL divergence, and training a classifier minimises it. The "free-energy" and predictive-coding theories of the brain are also written in this language.

5.7 Markov chains and sampling (optional, ≈*15 min*)

Finally, a glimpse of *how* Bayesian models are actually computed, closing the loop with the start of the session.

Definition

A **Markov chain** is a sequence of states in which the next state depends only on the current one, not the full history (the **Markov property**). Its dynamics are captured by a **transition matrix** of probabilities — a nice callback to linear algebra, since propagating the state distribution forward is repeated matrix multiplication.

Intuition

Posteriors in realistic models are too complicated to write down or integrate. So instead of computing them, we *sample* from them. **Markov chain Monte Carlo (MCMC)** builds a Markov chain that wanders through parameter space, spending time in each region in proportion to its posterior probability. The collected samples then approximate the posterior — histograms stand in for integrals. This is how most modern Bayesian cognitive models are fit in practice.

Why it matters for cognitive science

Beyond being a computational tool, sampling has been proposed as a model of cognition itself: perhaps the brain represents uncertainty by drawing samples, which would explain certain fluctuations and biases in perception and judgment. Markov models also describe sequential behaviour directly — state transitions in learning, sequences of eye movements, and spike-train dynamics.

Session recap

This session assembled the modeller's toolkit. *Bayesian inference* updates a full distribution over parameters (prior $\to$ posterior), summarised by the posterior mean or MAP. The *multivariate Gaussian* unites probability and linear algebra through the *covariance matrix*, whose eigenstructure is the PCA of Session 4. *Maximum likelihood* is the one principle behind most model fitting — and least squares is a special case. The *GLM* extends regression to binary and count data via link functions, giving psychometric and neural encoding models. *Model comparison* (cross-validation, AIC/BIC) defends against overfitting via the bias–variance tradeoff. *Information theory* measures uncertainty and shared information, and *Markov chains / sampling* show how Bayesian models are computed. Together with the four preceding sessions, students now hold the mathematical vocabulary of modern cognitive modeling, computational neuroscience, machine learning, and data analysis.